Department of Computer Science & Information Technology

CT-261 Database Management System

Complex Computing Activity (CCA) through Open-Ended Lab (OEL)

Student Name: ______________________ Student Roll No.: ______________________

Criteria	0 – Bad (0 Marks)	1 – Average (1 Mark)	2 – Good (2 Marks)
1. Domain Understanding & Problem Definition	Domain is unclear, trivial, or poorly defined with no real-world relevance.	Domain is partially defined with limited complexity and weak problem description.	Domain is clearly defined, complex, and represents a realistic real-world system with clear objectives.
2. Conceptual Design (ER Diagram)	ER diagram is missing, incorrect, or shows very few entities and relationships.	ER diagram includes basic entities and relationships but lacks clarity or completeness.	ER diagram clearly represents all required entities, relationships, and cardinalities accurately.
3. Normalization & Logical Design (3NF)	Tables are not normalized and show redundancy and anomalies.	Partial normalization is applied but some dependencies or anomalies remain.	Database schema is fully normalized up to 3NF with no insertion, deletion, or update anomalies.
4. Business Rules & Integrity Constraints	Business rules are missing or not enforced using constraints.	Some business rules are defined but weakly enforced.	Multiple meaningful business rules are clearly defined and enforced using primary keys, foreign keys, and constraints.
5. Analytical Queries & Innovation	Queries are missing or do not support decision-making; design shows no innovation.	Basic queries are provided with limited analytical value.	Meaningful analytical queries are included and the design shows innovative and creative application of DBMS principles.

Total Marks: ______/10____

Teacher's Signature: ____________

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Information Technology

Tech Club and Hackathon Management System

Spring 2026

Subject	Database Management System (DBMS)
Course Code	CT-261
Date	May 2026
Academic Batch	2024–2028
Instructor	Miss Mehar

GROUP MEMBERS

Name	Roll Number
Izma Warsia	CT-24311
Safa Khurram	CT-24310
Saniya Siddiqui	CT-24303
Ayesha	CT-24309

1. Executive Overview

1. Project Objective

The primary objective of this project is to design and implement a robust, scalable, and role-based Hackathon Management System. The system aims to automate the end-to-end lifecycle of a hackathon—from participant registration and team formation to event sponsorship, workshop coordination, and standardized judging. By providing distinct interfaces for Administrators, Mentors, Judges, and Participants, the system ensures a seamless flow of information and high data integrity through a normalized relational database.

2. Problem Statement

The organization of large-scale technical events, such as hackathons, is traditionally hindered by fragmented and manual processes that compromise both operational efficiency and data integrity. Conventional management methods frequently suffer from significant manual evaluation overheads, where tracking scores across diverse judging panels and numerous teams becomes highly susceptible to human error. Furthermore, inefficient data storage often leads to severe redundancy and update anomalies, particularly when managing overlapping participant, team, and workshop records. This lack of centralization creates a disconnected ecosystem where coordinating between sponsors, mentors, and participants becomes a logistical challenge. Additionally, the absence of robust, role-based access control often results in "Role Confusion," allowing for unauthorized data manipulation and a lack of accountability. These systemic inefficiencies necessitate a centralized, automated solution that can enforce strict competitive rules—such as project proposal approvals and team size constraints—while maintaining a single, normalized source of truth.

2.1. Real-World Context & Benchmark

To understand the gravity of these issues, we can look at existing platforms like Devpost or HackerEarth, which are global standards for hackathon management. While these platforms are powerful, many local educational institutions and private organizations face significant barriers when using them, such as:

- **Complexity & Cost:** Global platforms often have high licensing fees or complex configurations that are overkill for university-level seminars or departmental competitions.
- **The "University-Sponsor" Gap:** Most existing systems focus only on participants and judges. In a real-world university setting (like NED University), there is a critical need to link Departmental Seminars, Mentors, and Local Sponsors in one loop, which generic platforms often miss.
- **Rigid Rules:** Standard platforms may not allow hardcoded local constraints—such as our specific 3-member team limit or a mandatory Admin Approval for project ideas before a team is even considered "active."

2.2. The Inspiration (The Need for a Custom Solution)

This project was inspired by the logistical hurdles faced during large-scale technical fests where data is often scattered across Google Forms (for registration), WhatsApp Groups (for coordination), and Excel Sheets (for scoring).

The Real-World Problem: Imagine a hackathon with 50 teams. If a Judge enters a score in an Excel sheet that hasn't been updated with the latest "Admin-Approved" team list, the entire result becomes invalid. This "De-synchronized Data" is a real-world nightmare that our system solves by creating a Relational Single Source of Truth.

3. Business Requirements

To streamline high-stake technical competitions, the system is engineered to meet the following organizational mandates:

- **Operational Transparency:** Implementation of a centralized approval workflow to ensure only qualified project ideas enter the competition.
- **Data Reliability:** Maintenance of a "Single Source of Truth" by strictly adhering to relational integrity and normalization standards.
- **Resource Optimization:** Efficient allocation and tracking of event resources, including sponsorship funds and academic workshops.
- **Scalable Architecture:** A modular framework capable of handling concurrent multi-role interactions (Admin, Judges, Mentors, and Participants) without data collisions.
- **Objective Evaluation:** A standardized, bias-free scoring mechanism that automates results based on pre-defined technical criteria.

2. Solution Architecture

4. Proposed Solution

The project delivers a custom-built, end-to-end Hackathon Management Ecosystem designed on a relational foundation. The solution transitions from a monolithic, unorganized data structure into a 3NF-compliant MySQL schema comprising 11 interconnected entities. By leveraging the Streamlit framework, the system provides real-time data processing and interactive dashboards, ensuring that every user role—from an Admin approving a proposal to a Judge evaluating a submission—operates within a secure, high-performance environment.

4.1 Key Features

- **Admin Dashboard:** Oversight of hackathons, approval of team registrations, and management of sponsorship status.
- **Participant Dashboard:** Registration for events, team creation (capped at 3 members), and workshop enrollment.
- **Judge Dashboard:** Standardized scoring interface based on Technical, Innovative, and Executive criteria.
- **Mentor/Speaker Dashboard:** Coordination of specialized workshops and tracking of attendee lists.
- **Automated Results:** A dynamic results module that calculates winner standings through SQL Views.

5. Project Limitations & Future Enhancements

5.1 Current System Limitations

While the system is highly optimized for relational data management, the current version has the following constraints:

- **Static Multimedia Support:** The system currently does not support the direct upload or streaming of large video files (e.g., project demos); it relies on external GitHub and URL links to maintain database performance.
- **Manual Notification Workflow:** Status updates (like registration approval or scoring) require users to log in to their respective dashboards as there is no automated email or SMS notification trigger integrated yet.
- **Sequential Evaluation:** The system is designed for a single-event structure. Handling multiple concurrent hackathons with different scoring rubrics simultaneously may require further schema expansion.
- **Lack of Real-time Chat:** There is no built-in communication channel between mentors and participants, requiring teams to use external platforms for technical guidance.

5.2 Future Enhancements

To evolve this platform into a comprehensive event management ecosystem, the following features are proposed for future development:

- **AI-Powered Team Matching:** Implementation of an AI module to suggest potential team members to individual participants based on complementary skill sets (e.g., linking a Backend Developer with a UI/UX Designer).
- **Automated Communication Engine:** Integration with SMTP servers or APIs like Twilio to send automated real-time notifications for approval statuses, deadline reminders, and final results.
- **Live Judging Leaderboard:** A public-facing, real-time leaderboard utilizing Plotly or Streamlit's live update features to show shifting team rankings as judges submit their scores.
- **Integrated IDE & Sandbox:** Providing a lightweight, browser-based code editor or a sandbox environment for participants to submit "Proof of Concept" (PoC) snippets directly within the platform.
- **Certificate Generation:** An automated module for the Admin to generate and issue digital "Certificates of Participation" and "Winner Certificates" using data dynamically pulled from the MySQL backend.
- **Payment Gateway Integration:** For paid events, integrating a secure payment gateway (like Stripe or PayPal) to automate registration fee collection and sponsor transaction tracking.

3. Conceptual Model

6. Entities Description

- **User:** A central entity representing all actors (Admin, Participant, Mentor, Judge) with unique credentials and roles.
- **Hackathon:** The core event entity containing details like domain, registration deadlines, and prize pools.
- **Team:** A group of participants (max 3) formed to compete in a specific hackathon.
- **Project:** The digital solution/idea submitted by a team for evaluation, linked to a GitHub repository.
- **Workshop:** Educational sessions organized within a hackathon to guide participants.
- **Sponsor:** Organizations providing financial backing or resources for the hackathon events.
- **Judge:** Specialized users responsible for evaluating projects based on technical and innovative criteria.
- **Scores:** An associative entity that stores evaluation metrics for a project within a specific hackathon.

7. Entity-Relationship Logic (Cardinality)

7.1 Administrative Relationships

- **Admin manages Hackathons (1:N):** One Admin can create and oversee multiple hackathons, but each hackathon is managed under the primary admin's supervision.
- **Admin manages Workshops (1:N):** One Admin coordinates multiple workshops across different hackathons.
- **Admin manages Sponsors (1:N):** An Admin handles the registration and status tracking of multiple sponsors.

7.2 Event & Participation Relationships

- **Hackathon to Workshops (1:N):** One hackathon can host multiple workshops, but a specific workshop belongs strictly to one parent hackathon.
- **Sponsor funds Hackathons (M:N):** A single sponsor can fund multiple hackathons, and one hackathon can be supported by multiple sponsors. This requires a junction table (e.g., Sponsor_Hackathon).
- **Participant attends Workshop (M:N) with Constraint:** Logic: While many participants can attend many workshops, the system enforces a Business Rule: A participant can only enroll in a workshop that belongs to the hackathon they are registered in. Technically, this is an M:N relationship resolved via a Workshop_Attendees table.

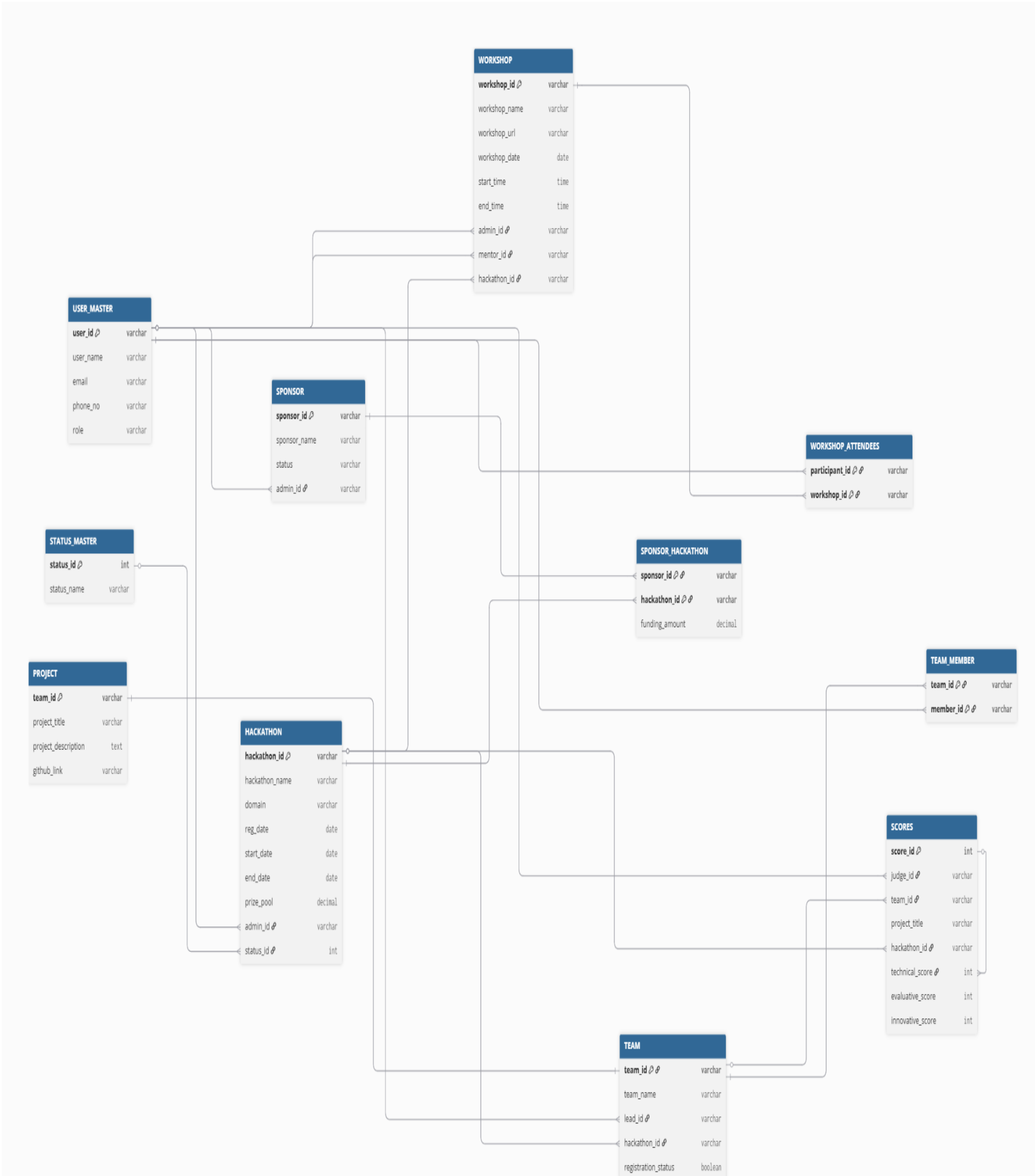
7.3 Team & Project Relationships

- **Participant Joins Team (M:N) with Constraint:** A participant can belong to teams, but the system strictly enforces a 3-member limit per team.
- **Team Lead leads Team (1:N):** One member is designated as the 'Lead' who manages the other 2 members and holds submission authority.
- **Team registers in Hackathon (M:1):** Many teams register for a single hackathon.
- **Team submits Project (1:1):** Each team submits exactly one project for a specific hackathon.

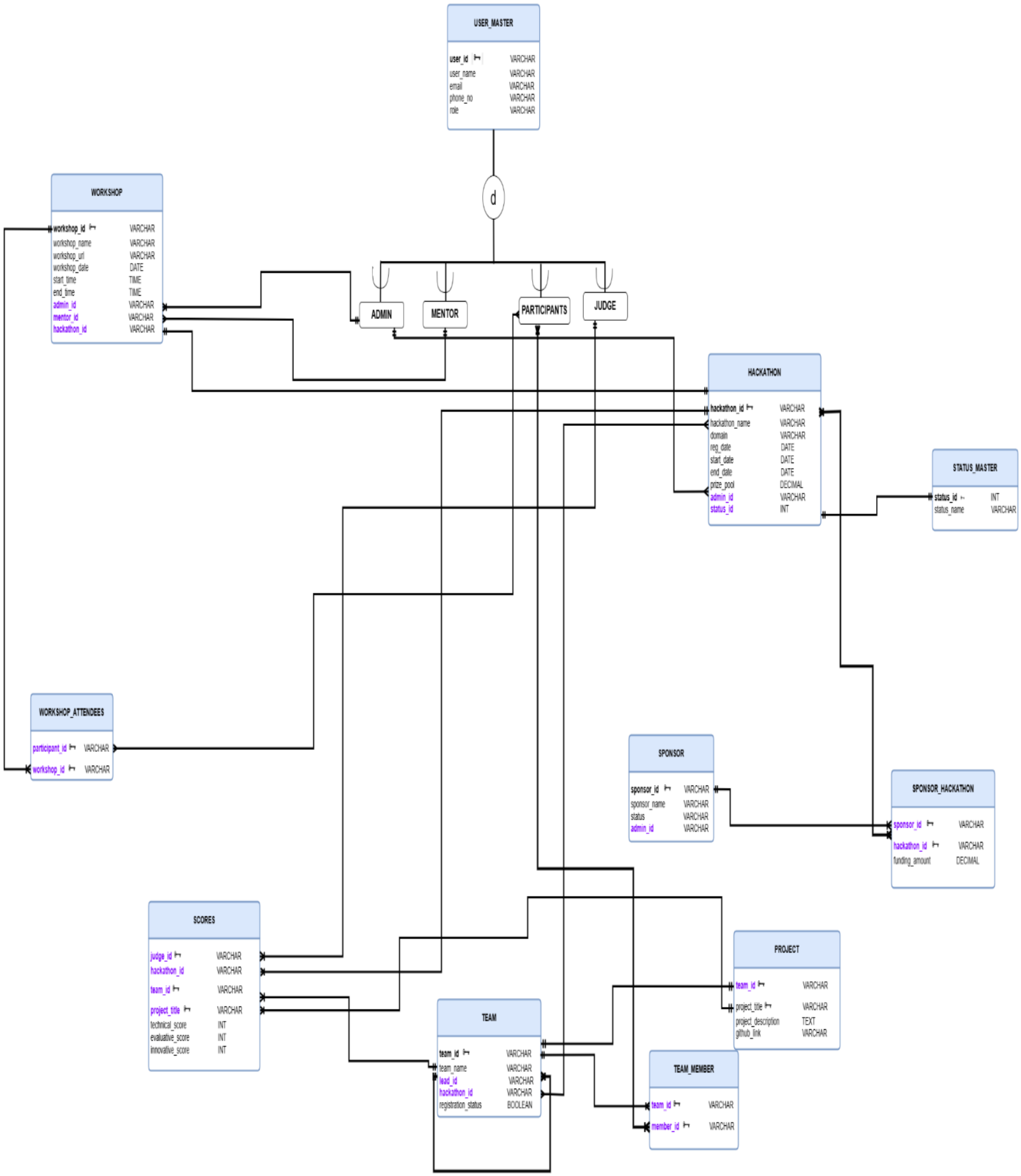
7.4 Evaluation (The Ternary Relationship)

- **Scores (Associative Entity):** This is a Ternary Relationship between Judge, Project, and Hackathon.
- **Logic:** A Score record is only created when a specific Judge evaluates a specific Project within the context of a specific Hackathon. It acts as a junction point for these three entities.

8. ERD DIAGRAM:



9. EERD DIAGRAM:



4. Logical Model

10. UN-NORMALIZED DATASET:

User_name	User_contact	User_role
Ayesha Khan	0347-1212121 | user66@email.com	Participant
Ayesha Khan	0347-1212121 | user66@email.com	Participant
Ayesha Khan	0347-1212121 | user66@email.com	Participant
Ahmed Raza	0300-1112223 | org-ahmed1@gmail.com	Admin Staff
Mustafa Kamal	0345-8889990 | user12@email.com	Participant
Mustafa Kamal	0345-8889990 | user12@email.com	Participant
Mustafa Kamal	0345-8889990 | user12@email.com	Participant
Engr. Waqas Ahmed	0345-1122334 | j-user454@gmail.com	Judge
Engr. Waqas Ahmed	0345-1122334 | j-user454@gmail.com	Judge
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Mustafa Qadri	0345-0001112 | mentor.mustafa@email.com	Mentor
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Yasir Nawaz	0344-2223334 | mentor.yasir@email.com	Mentor
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Marium Khan	0345-0001112 | org-mari4@gmail.com	Admin Staff
Yasir Hussain	0344-9990001 | user994@email.com	Participant
Yasir Hussain	0344-9990001 | user994@email.com	Participant
Yasir Hussain	0344-9990001 | user994@email.com	Participant
Yasir Nawaz	0344-2223334 | mentor.yasir@email.com	Mentor

Continue….

Team_name	Team_members	Project_name	Project_description	Github-link
SecureSquad	Hamza|Ayesha|Osama	ShieldNet	Network Threat Detector	github.com/sec-s
SecureSquad	Hamza|Ayesha|Osama	ShieldNet	Network Threat Detector	github.com/sec-s
SecureSquad	Hamza|Ayesha|Osama	ShieldNet	Network Threat Detector	github.com/sec-s
ByteBusters	Bilal|Saira|Rohan	HealthTrack	AI Health Monitoring	github.com/byte-h
ChainGang	Bilal|Hina|Mustafa	CryptoPay	Blockchain Payment App	github.com/chain-p
ChainGang	Bilal|Hina|Mustafa	CryptoPay	Blockchain Payment App	github.com/chain-p
ChainGang	Bilal|Hina|Mustafa	CryptoPay	Blockchain Payment App	github.com/chain-p
ByteBusters	Bilal|Saira|Rohan	HealthTrack	AI Health Monitoring	github.com/byte-h
ByteBusters	Bilal|Saira|Rohan	HealthTrack	AI Health Monitoring	github.com/byte-h
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
DeepMinders	Bilal|Umar|Babar	BrainScan	Neural Image Processing	github.com/deep-m
WebWarriors	Saira|Yasir|Mehak	SiteSwift	Fast Web Rendering	github.com/web-w
WebWarriors	Saira|Yasir|Mehak	SiteSwift	Fast Web Rendering	github.com/web-w
WebWarriors	Saira|Yasir|Mehak	SiteSwift	Fast Web Rendering	github.com/web-w

Hackathon_title	Hackathon_domain	Hackathon_stat	Reg-deadline	Commencement_d	End_date	Prize_pool
Query Quest	Data Science	Evaluated	1/28/2026	2/1/2026	2/3/2026	3000
Algo Artist	AI & ML	Evaluated	2/10/2026	2/14/2026	2/16/2026	3500
Kube Kraft	Cloud Computing	Ongoing	7/1/2026	7/5/2026	7/7/2026	6000
Neural Network Ninja	AI & ML	Ongoing	5/10/2026	5/14/2026	5/16/2026	5000
Kotlin King	Mobile App Dev	Ongoing	6/28/2026	7/2/2026	7/4/2026	6000
Solana Sprint	Blockchain	Upcoming	9/10/2026	9/14/2026	9/16/2026	7000
Docker Dash	Cloud Computing	Upcoming	9/25/2026	9/29/2026	10/1/2026	3500
ChatBot Championship	AI & ML	Evaluated	3/1/2026	3/5/2026	3/7/2026	3000
ChatBot Championship	AI & ML	Evaluated	3/1/2026	3/5/2026	3/7/2026	3000
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Predictive Pros	AI & ML	Ongoing	7/20/2026	7/24/2026	7/26/2026	5500
Malware Mania	Cybersecurity	Evaluated	2/22/2026	2/26/2026	2/28/2026	4000
Shield Sprint	Cybersecurity	Ongoing	4/12/2026	4/16/2026	4/18/2026	5000
Encryption Elite	Cybersecurity	Upcoming	12/5/2026	12/9/2026	12/11/2026	5500

Sponsor_name	Sponsor_status	Sponsor_amount
Snowflake	Accepted	4000
Anthropic	Accepted	4500
DigitalOcean	Accepted	7500
NVIDIA	Accepted	6500
JetBrains	Accepted	7500
Solana Labs	Pending	9000
RedHat	Pending	4500
Meta	Accepted	7000
Meta	Accepted	7000
Meta	Accepted	7000
Meta	Accepted	7000
Meta	Accepted	7000
Meta	Accepted	7000
Palo Alto Networks	Accepted	5000
CrowdStrike	Accepted	7000
Fortinet	Pending	6500

Workshop_title	Workshop_URL	Date	Time
Modern React.js	https://react-dev.com/hooks	5/16/2026	10:00 AM | 12:00 PM
Intro to Neural Networks	https://learn-ai.io/nn-101	5/12/2026	10:00 AM | 12:00 PM
Node.js Backend	https://backend-hub.io/node	8/7/2026	02:00 PM | 04:00 PM
Intro to Neural Networks	https://learn-ai.io/nn-101	5/12/2026	10:00 AM | 12:00 PM
NFT Minting Guide	https://nft-school.xyz/mint	5/27/2026	04:00 PM | 06:00 PM
Modern React.js	https://react-dev.com/hooks	5/16/2026	10:00 AM | 12:00 PM
Cryptography Essentials	https://crypto-safes.edu/101	12/7/2026	11:00 AM | 01:00 PM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
Web3 Identity (DID)	https://web3-auth.com/did	9/12/2026	10:00 AM | 12:00 PM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
PyTorch for Beginners	https://pytorch-labs.edu/intro	7/22/2026	09:00 AM | 11:00 AM
Intro to Neural Networks	https://learn-ai.io/nn-101	5/12/2026	10:00 AM | 12:00 PM
OWASP Top 10	https://security-code.org/owasp	2/24/2026	09:00 AM | 11:00 AM
OWASP Top 10	https://security-code.org/owasp	2/24/2026	09:00 AM | 11:00 AM
Web3 Identity (DID)	https://web3-auth.com/did	9/12/2026	10:00 AM | 12:00 PM

Technical_score	Innovative_score	Executive_score	Final_score
7	6	8	12th Rank
7	6	8	12th Rank
6	7	7	10th Rank
6	7	7	10th Rank
9	9	8	2nd Rank

11. Normalization Process

11.1 First Normal Form (1NF):

It ensures that each table cell contains only a single atomic value and there are no repeating groups or multi-valued attributes.

COMPOSITE_PK: {user_email, hacakthon_title, workshop_title, team_name, team_members}

User_name	User_phone	User_email	User_role
Amna Sheikh	0301-7776665	user992@email.com	Participant
Amna Sheikh	0301-7776665	user992@email.com	Participant
Amna Sheikh	0301-7776665	user992@email.com	Participant
Amna Sheikh	0301-7776665	user992@email.com	Participant
Amna Sheikh	0301-7776665	user992@email.com	Participant
Amna Sheikh	0301-7776665	user992@email.com	Participant
Nimra Ali	0331-4445556	user77@email.com	Participant
Nimra Ali	0331-4445556	user77@email.com	Participant
Nimra Ali	0331-4445556	user77@email.com	Participant
Bilal Ashraf	0313-6665554	j-user808@gmail.com	Judge
Bilal Ashraf	0313-6665554	j-user808@gmail.com	Judge
Bilal Ashraf	0313-6665554	j-user808@gmail.com	Judge
Maya Khan	0323-7778889	j-user909@gmail.com	Judge
Maya Khan	0323-7778889	j-user909@gmail.com	Judge
Maya Khan	0323-7778889	j-user909@gmail.com	Judge
Ahmed Raza	0300-1112223	org-ahmed1@gmail.com	Admin Staff
Sanam Baloch	0302-7778889	org-sanam17@gmail.com	Admin Staff
Imran Abbas	0329-4445556	org-imran16@gmail.com	Admin Staff
Mahira Hafeez	0334-5544667	mentor.mahira@email.com	Mentor
Momina Mustehsan	0321-7788221	mentor.momina@email.com	Mentor
Mustafa Qadri	0345-0001112	mentor.mustafa@email.com	Mentor
Naseem Jaffer	0339-6677667	mentor.naseem@email.com	Mentor
Osama Khalid	0305-9988776	mentor.osama@email.com	Mentor

Continued…..

Team_name	Team_members	Project_name	Project_description	Github-link
AppArtisans	Sana	SwiftOrder	Food Delivery System	github.com/app-a
DataWizards	Fahad	StatGen	Automated Data Insights	github.com/data-s
AppArtisans	Tariq	SwiftOrder	Food Delivery System	github.com/app-a
AppArtisans	Tariq	SwiftOrder	Food Delivery System	github.com/app-a
AppArtisans	Sana	SwiftOrder	Food Delivery System	github.com/app-a
DataWizards	Nimra	StatGen	Automated Data Insights	github.com/data-s
DataWizards	Amna	StatGen	Automated Data Insights	github.com/data-s
DataWizards	Nimra	StatGen	Automated Data Insights	github.com/data-s
ChainGang	Bilal	CryptoPay	Blockchain Payment App	github.com/chain-p
LogicLords	Amna	PureLogic	Clean Code Engine	github.com/logic-l
AppArtisans	Sana	SwiftOrder	Food Delivery System	github.com/app-a
AppArtisans	Sana	SwiftOrder	Food Delivery System	github.com/app-a
AppArtisans	Kiran	SwiftOrder	Food Delivery System	github.com/app-a
DataWizards	Fahad	StatGen	Automated Data Insights	github.com/data-s
DataWizards	Amna	StatGen	Automated Data Insights	github.com/data-s
ChainGang	Mustafa	CryptoPay	Blockchain Payment App	github.com/chain-p
LogicLords	Asma	PureLogic	Clean Code Engine	github.com/logic-l
DataWizards	Nimra	StatGen	Automated Data Insights	github.com/data-s
AppArtisans	Kiran	SwiftOrder	Food Delivery System	github.com/app-a
DataWizards	Amna	StatGen	Automated Data Insights	github.com/data-s
LogicLords	Fawad	PureLogic	Clean Code Engine	github.com/logic-l
DataWizards	Fahad	StatGen	Automated Data Insights	github.com/data-s
WebWarriors	Saira	SiteSwift	Fast Web Rendering	github.com/web-w

Hackathon_title	Hackathon_domain	Hackathon_status	Reg-deadline	Commencement_date	End_date	Prize_pool	Sponsor_name	Sponsor_status
Ansible Ace	DevOps	Evaluated	2/5/2026	2/9/2026	2/11/2026	5000	Adobe	Accepted
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Adobe	Pending
Ansible Ace	DevOps	Evaluated	2/5/2026	2/9/2026	2/11/2026	5000	Anthropic	Accepted
Flutter Flow	Mobile App Dev	Upcoming	10/10/2026	10/14/2026	10/16/2026	4500	Anthropic	Pending
Flutter Flow	Mobile App Dev	Upcoming	10/10/2026	10/14/2026	10/16/2026	4500	Anthropic	Pending
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Anthropic	Pending
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Anthropic	Pending
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Anthropic	Pending
Chain Reaction	Blockchain	Ongoing	4/5/2026	4/9/2026	4/11/2026	6000	Apple	Accepted
Query Quest	Data Science	Evaluated	1/28/2026	2/1/2026	2/3/2026	3000	Apple	Accepted
Flutter Flow	Mobile App Dev	Upcoming	10/10/2026	10/14/2026	10/16/2026	4500	Apple	Pending
React Race	Web Development	Ongoing	5/14/2026	5/18/2026	5/20/2026	3000	Apple	Accepted
Flutter Flow	Mobile App Dev	Upcoming	10/10/2026	10/14/2026	10/16/2026	4500	Apple	Pending
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Apple	Pending
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Apple	Pending
Chain Reaction	Blockchain	Ongoing	4/5/2026	4/9/2026	4/11/2026	6000	AWS	Accepted
Query Quest	Data Science	Evaluated	1/28/2026	2/1/2026	2/3/2026	3000	AWS	Accepted
Pandas Play	Data Science	Ongoing	4/20/2026	4/24/2026	4/26/2026	4000	AWS	Accepted
React Race	Web Development	Ongoing	5/14/2026	5/18/2026	5/20/2026	3000	AWS	Accepted
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	AWS	Pending
Query Quest	Data Science	Evaluated	1/28/2026	2/1/2026	2/3/2026	3000	Azure	Accepted
Pandas Play	Data Science	Ongoing	4/20/2026	4/24/2026	4/26/2026	4000	Azure	Accepted
React Race	Web Development	Ongoing	5/14/2026	5/18/2026	5/20/2026	3000	Azure	Accepted
React Race	Web Development	Ongoing	5/14/2026	5/18/2026	5/20/2026	3000	Azure	Accepted
Deep Dive Vision	AI & ML	Upcoming	6/15/2026	6/19/2026	6/21/2026	4500	Azure	Pending

Workshop_title	Workshop_URL	Date	Workshop_start_time	Workshop_end_time
AWS Essentials	https://aws-cloud.com/basics	5/4/2026	10:00:00 AM	12:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
AWS Essentials	https://aws-cloud.com/basics	5/4/2026	10:00:00 AM	12:00:00 PM
React Native Basics	https://rn-hub.org/start	1/14/2026	11:00:00 AM	1:00:00 PM
React Native Basics	https://rn-hub.org/start	1/14/2026	11:00:00 AM	1:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
NFT Minting Guide	https://nft-school.xyz/mint	5/27/2026	4:00:00 PM	6:00:00 PM
Solidity Smart Contracts	https://eth-build.io/solidity	4/7/2026	1:00:00 PM	3:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
Flutter Framework	https://flutter-dev.io/intro	5/24/2026	10:00:00 AM	1:00:00 PM
Tailwind CSS	https://style-pro.xyz/tailwind	7/17/2026	9:00:00 AM	11:00:00 AM
React Native Basics	https://rn-hub.org/start	1/14/2026	11:00:00 AM	1:00:00 PM
NFT Minting Guide	https://nft-school.xyz/mint	5/27/2026	4:00:00 PM	6:00:00 PM
NFT Minting Guide	https://nft-school.xyz/mint	5/27/2026	4:00:00 PM	6:00:00 PM
Solidity Smart Contracts	https://eth-build.io/solidity	4/7/2026	1:00:00 PM	3:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
Python for Data Science	https://py-data.org/course	4/22/2026	10:00:00 AM	1:00:00 PM
Tailwind CSS	https://style-pro.xyz/tailwind	7/17/2026	9:00:00 AM	11:00:00 AM
Node.js Backend	https://backend-hub.io/node	8/7/2026	2:00:00 PM	4:00:00 PM
Advanced SQL Queries	https://sql-pro.io/adv	1/30/2026	11:00:00 AM	1:00:00 PM
Python for Data Science	https://py-data.org/course	4/22/2026	10:00:00 AM	1:00:00 PM
Modern React.js	https://react-dev.com/hooks	5/16/2026	10:00:00 AM	12:00:00 PM
Tailwind CSS	https://style-pro.xyz/tailwind	7/17/2026	9:00:00 AM	11:00:00 AM
Node.js Backend	https://backend-hub.io/node	8/7/2026	2:00:00 PM	4:00:00 PM

hackathon_id	hackathon_name	domain	reg_date	start_date	end_date	prize_pool	admin_id	status_name	judge_id
H-001	Query Quest	Web development	28/05/2026	01/06/2026	03/06/2026	3000	U-001	Upcoming	U-026
H-002	Malware Mania	Web development	22/02/2026	26/02/2026	28/02/2026	4000	U-003	Evaluated	U-026
H-003	Ansible Ace	DevOps	05/06/2026	09/06/2026	11/06/2026	5000	U-021	Upcoming	U-036
H-004	Angular Arena	Web development	25/05/2026	01/06/2026	03/06/2026	4000	U-022	Upcoming	U-026
H-005	DeFi Derby	Blockchain	20/05/2026	24/05/2026	26/05/2026	5000	U-024	Upcoming	U-038
H-006	ChatBot Championship	AI & ML	01/03/2026	05/03/2026	07/03/2026	3000	U-025	Evaluated	U-030
H-007	Neural Network Ninja	AI & ML	10/04/2026	14/04/2026	16/04/2026	5000	U-002	Evaluated	U-029
H-008	NFT Knights	Blockchain	27/04/2026	01/05/2026	03/05/2026	4000	U-004	Ongoing	U-039
H-009	Jenkins Jive	DevOps	27/04/2026	01/05/2026	03/05/2026	4500	U-007	Ongoing	U-040
H-010	Chain Reaction	Blockchain	05/04/2026	09/04/2026	11/04/2026	6000	U-009	Evaluated	U-034
H-011	React Race	Web Development	27/04/2026	01/05/2026	03/05/2026	3000	U-010	Ongoing	U-041
H-012	Predictive Pros	AI & ML	27/04/2026	01/05/2026	03/05/2026	5500	U-013	Ongoing	U-042
H-013	Full Stack Fever	Web Development	27/04/2026	01/05/2026	03/05/2026	5000	U-014	Ongoing	U-027
H-014	Cloud Castles	Cloud Computing	02/04/2026	06/04/2026	08/04/2026	5000	U-015	Evaluated	U-028
H-015	Pipeline Pro	DevOps	27/04/2026	01/05/2026	03/05/2026	6500	U-016	Ongoing	U-036
H-016	Pandas Play	Data Science	20/04/2026	24/04/2026	26/04/2026	4000	U-018	Evaluated	U-032
H-017	Swift Safari	Mobile App Dev	22/04/2026	26/04/2026	28/04/2026	5000	U-019	Evaluated	U-034
H-018	Shield Sprint	Cybersecurity	12/04/2026	16/04/2026	18/04/2026	5000	U-020	Evaluated	U-026
H-019	Serverless Summit	Cloud Computing	20/03/2026	24/03/2026	26/03/2026	4500	U-005	Evaluated	U-035
H-020	Deep Dive Vision	AI & ML	15/06/2026	19/06/2026	21/06/2026	4500	U-006	Upcoming	U-037
H-021	Git Guide	DevOps	18/09/2026	22/09/2026	24/09/2026	3000	U-008	Upcoming	U-038
H-022	Solana Sprint	Blockchain	10/04/2026	14/04/2026	16/04/2026	7000	U-011	Evaluated	U-043
H-023	Pentest Party	Cybersecurity	01/10/2026	05/10/2026	07/10/2026	6500	U-012	Upcoming	U-039
H-024	Flutter Flow	Mobile App Dev	10/02/2026	14/02/2026	16/02/2026	4500	U-017	Evaluated	U-031

Technical_score	Innovative_score	Executive_score	Final_sum
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
10	10	9	29
9	10	10	29
9	10	10	29
9	10	10	29
9	10	10	29
9	10	10	29
9	10	10	29
9	9	10	28
9	9	10	28
9	9	10	28
9	9	10	28
9	9	10	28
9	9	10	28
10	9	9	28
10	9	9	28
10	9	9	28
9	10	9	28

11.1.1 ANOMALIES ANALYSIS:

This is a analyzes of 1NF dataset for identifying common database anomalies:

- Insertion Anomaly
- Update Anomaly
- Deletion Anomaly

The dataset contains information about users, teams, projects, sponsors, workshops, and hackathons in a single table, which leads to redundancy and inconsistency issues.

Insertion Anomaly:

We cannot insert a new entity without inserting unnecessary data.

Example:

An insertion anomaly exists where a new workshop cannot be recorded in the system without first being associated with a pre-existing hackathon. This dependency prevents the staging of independent workshop data prior to the formal establishment of a hackathon entity.

Query:

```
SELECT * FROM workshop_attendees;

INSERT INTO `1nf_anomilies`
(Workshop_title, Workshop_URL)
VALUES ('AI Bootcamp', 'github.com/ai-bootcamp');
```

Issue:

We are forced to provide user, team, and project data which may not exist yet.

Update Anomaly

Problem:

Same data is repeated in multiple rows, so updating requires multiple changes.

Example:

An update anomaly was identified where modifications to a single user's record failed to propagate across the system. This inconsistency indicates a lack of normalization, resulting in redundant data that remains in its previous state.

Query:

```
User phone number is repeated:
UPDATE `1nf_anomilies`
SET User_phone = '0300-1112223'
WHERE User_name = 'Kiran Mazumdar';
```

Issue:

This user appears in multiple rows → missing update in one row causes inconsistency.

Deletion Anomaly

Problem:

Deleting a row may remove important unrelated data.

Example:

A deletion anomaly has been identified where the removal of a specific record results in the unintended loss of unrelated data. For example, deleting a Hackathon record inadvertently wipes all associated Workshop and Participant history, leading to an irrecoverable loss of institutional data.

Query:

```
DELETE FROM `1nf_anomilies`
WHERE Project_name = 'StatGen';
```

Issue:

If this was the only record for a workshop/sponsor, that information is also lost.

Real Impact of Anomalies

- Data redundancy
- Inconsistent updates
- Loss of important information
- Poor database efficiency

11.2 Second Normal Form (2NF):

It requires the table to be in 1NF and ensures that all non-key attributes are fully functionally dependent on the entire primary key, eliminating partial dependencies.

Partial Dependency:

User Info

User_email → (User_name, User_phone, User_role)

Hackathon Details

Hackathon_title → (Hackathon_domain, Reg_deadline, Commencement_date, End_date, Prize_pool, Sponsor_amount)

Workshop Details

Workshop_title → (Workshop_URL, Date, Workshop_start_time, Workshop_end_time)

Team & Project

Team_name → (Project_name, Project_description, Registration_status)

Scores Info

(Hackathon_title, Team_name, Team_members) → (Technical_score, Innovative_score, Executive_score, Final_score)

USER_MASTER

PK

user_id	user_name	email	phone_no	role
U-001	Ahad Raza Mir	org-ahad20@gmail.com	0334-5556667	Admin Staff
U-002	Ahmed Raza	org-ahmed1@gmail.com	0300-1112223	Admin Staff
U-003	Bilal Saeed	org-bilal12@gmail.com	0328-2223334	Admin Staff
U-004	Bushra Ansari	org-bush9@gmail.com	0332-4445556	Admin Staff
U-005	Fahad Mustafa	org-fahad14@gmail.com	0340-8889990	Admin Staff
U-006	Fatima Sohail	org-fatima2@gmail.com	0321-4445556	Admin Staff
U-007	Hamza Ali	org-hamza18@gmail.com	0313-0001112	Admin Staff
U-008	Hania Aamir	org-hania23@gmail.com	0314-4445556	Admin Staff
U-009	Hassan Ali	org-hassan6@gmail.com	0305-5556667	Admin Staff
U-010	Imran Abbas	org-imran16@gmail.com	0329-4445556	Admin Staff
U-011	Iqra Aziz	org-iqra7@gmail.com	0318-8889990	Admin Staff
U-012	Kubra Khan	org-kub11@gmail.com	0309-0001112	Admin Staff
U-013	Marium Khan	org-mari4@gmail.com	0345-0001112	Admin Staff
U-014	Maya Ali	org-maya21@gmail.com	0342-7778889	Admin Staff
U-015	Mehwish Hayat	org-meh13@gmail.com	0339-5556667	Admin Staff
U-016	Momina Mustehsan	org-mom25@gmail.com	0335-0001112	Admin Staff
U-017	Osman Khalid	org-osman22@gmail.com	0303-1112223	Admin Staff
U-018	Sajal Ali	org-sajal15@gmail.com	0311-1112223	Admin Staff
U-019	Sanam Baloch	org-sanam17@gmail.com	0302-7778889	Admin Staff
U-020	Sheheryar Khan	org-sheh10@gmail.com	0341-7778889	Admin Staff
U-021	Shehzad Roy	org-roy24@gmail.com	0325-7778889	Admin Staff
U-022	Syra Yousuf	org-syra19@gmail.com	0323-2223334	Admin Staff
U-023	Usman Tariq	org-usman5@gmail.com	0312-2223334	Admin Staff
U-024	Waleed Khan	org-waleed8@gmail.com	0324-1112223	Admin Staff
U-025	Zeeshan Mani	org-zee3@gmail.com	0333-7778889	Admin Staff
U-026	Adnan Sami	j-user505@gmail.com	0318-1234567	Judge
U-027	Ayesha Omer	j-user404@gmail.com	0340-4445556	Judge
U-028	Bilal Ashraf	j-user808@gmail.com	0313-6665554	Judge
U-029	Dr. Arshad Khan	j-user786@gmail.com	0300-9988776	Judge
U-030	Engr. Waqas Ahmed	j-user454@gmail.com	0345-1122334	Judge
U-031	Gohar Rasheed	j-user111@gmail.com	0334-1212121	Judge
U-032	Junaid Khan	j-user303@gmail.com	0339-1112223	Judge
U-033	Kamran Akmal	j-user707@gmail.com	0332-5556667	Judge
U-034	Maya Khan	j-user909@gmail.com	0323-7778889	Judge
U-035	Misbah-ul-Haq	j-user101@gmail.com	0309-8889990	Judge
U-036	Noman Ijaz	j-user333@gmail.com	0303-5656565	Judge
U-037	Prof. Samina Ali	j-user212@gmail.com	0321-5544332	Judge
U-038	Rameez Raja	j-user777@gmail.com	0343-4455667	Judge
U-039	Sarmad Khoosat	j-user222@gmail.com	0342-3434343	Judge
U-040	Shoaib Malik	j-user555@gmail.com	0325-9090909	Judge
U-041	Wasim Akram	j-user666@gmail.com	0335-1122334	Judge
U-042	Younis Khan	j-user202@gmail.com	0328-7776665	Judge
U-043	Zubair Siddiqui	j-user121@gmail.com	0312-3344556	Judge
U-044	Abida Rahid	mentor.abida@email.com	0314-3322110	Mentor
U-045	Ali Sethi	mentor.ali@email.com	0306-4455661	Mentor
U-046	Amna Ali	mentor.amna@email.com	0301-5554443	Mentor
U-047	Asma Shirazi	mentor.asma@email.com	0313-7766554	Mentor
U-048	Atif Siddiqui	mentor.atif@email.com	0342-9988112	Mentor
U-049	Babar Ali	mentor.babar@email.com	0309-2233223	Mentor
U-050	Dr. Bilal Shah	mentor.bilal@email.com	0300-9998887	Mentor
U-051	Fahad Mustafa	mentor.fahad@email.com	0322-6667778	Mentor
U-052	Fawad Chaudhry	mentor.fawad@email.com	0323-1122445	Mentor
U-053	Hamza Ali	mentor.hamza@email.com	0336-1122112	Mentor
U-054	Haris Sohail	mentor.haris@email.com	0329-3344334	Mentor
U-055	Hina Mani	mentor.hina@email.com	0315-9990001	Mentor
U-056	Mahira Hafeez	mentor.mahira@email.com	0334-5544667	Mentor
U-057	Momina Mustehsan	mentor.momina@email.com	0321-7788221	Mentor
U-058	Mustafa Qadri	mentor.mustafa@email.com	0345-0001112	Mentor
U-059	Naseem Jaffer	mentor.naseem@email.com	0339-6677667	Mentor

U-060	Osama Khalid	mentor.osama@email.com	0305-9988776	Mentor
U-061	Prof. Rohan	mentor.rohan@email.com	0333-4443332	Mentor
U-062	Saira Khan	mentor.saira@email.com	0321-1112223	Mentor
U-063	Shadab Lodhi	mentor.shadab@email.com	0340-9900990	Mentor
U-064	Shaheen Shah	mentor.shaheen@email.com	0328-5544332	Mentor
U-065	Tariq Amin	mentor.tariq@email.com	0324-6655443	Mentor
U-066	Yasir Nawaz	mentor.yasir@email.com	0344-2223334	Mentor
U-067	Zainab Omar	mentor.zainab@email.com	0320-9876789	Mentor
U-068	Amna Sheikh	user992@email.com	0301-7776665	Participant
U-069	Ayesha Khan	user66@email.com	0347-1212121	Participant
U-070	Bilal Shah	user1@email.com	0300-1234567	Participant
U-071	Fahad Khan	user88@email.com	0322-1112223	Participant
U-072	Hamza Ahmed	user55@email.com	0336-7778889	Participant
U-073	Hina Dilpazeer	user50@email.com	0315-2223334	Participant
U-074	Kiran Mazumdar	user11@email.com	0332-9090909	Participant
U-075	Mehak Malik	user22@email.com	0300-4443332	Participant
U-076	Mustafa Kamal	user12@email.com	0345-8889990	Participant
U-077	Nimra Ali	user77@email.com	0331-4445556	Participant
U-078	Osama Bin	user78@email.com	0305-3434343	Participant
U-079	Rohan Sheikh	user99@email.com	0333-5556667	Participant
U-080	Saad Salman	user33@email.com	0312-5554443	Participant
U-081	Saira Bano	user10@email.com	0321-9876543	Participant
U-082	Sana Javed	user89@email.com	0318-5656565	Participant
U-083	Tariq Aziz	user90@email.com	0324-7878787	Participant
U-084	Yasir Hussain	user994@email.com	0344-9990001	Participant
U-085	Zainab Abbas	user44@email.com	0320-6665554	Participant
U-086	Safa Khurram	safa@gmail.com	0326-4231789	Participant

TEAM:

PK **FK** **FK**

team_id	team_name	lead_id	hackathon_id	registration_status
T-001	DataWizards	U-068	H-016	1
T-002	SecureSquad	U-069	H-018	1
T-003	ByteBusters	U-070	H-006	1
T-004	ChainGang	U-073	H-010	1
T-005	AppArtisans	U-074	H-017	1
T-006	WebWarriors	U-075	H-003	1
T-007	CloudKings	U-079	H-019	1
T-008	meshing	U-086	H-023	1
T-009	NeuralKnights	U-068	H-016	1
T-010	VectorVoyagers	U-069	H-016	1
T-011	CipherSync	U-070	H-018	1
T-012	ZeroTrustZenith	U-071	H-018	1
T-013	PromptProphets	U-072	H-006	1
T-014	InsightInquisitors	U-073	H-006	1
T-015	LogicLoom	U-074	H-006	1
T-016	HashHustlers	U-075	H-010	1
T-017	BlockBastions	U-076	H-010	1
T-018	BinaryBridges	U-077	H-010	1
T-019	SwiftSynthetics	U-078	H-017	1
T-020	ServerlessSamurais	U-080	H-019	1
T-021	DataDrifters	U-082	H-008	1
T-022	AlgoAces	U-083	H-009	1
T-023	CyberCrusaders	U-084	H-008	1
T-024	ShieldShifters	U-085	H-009	1
T-025	CloudComets	U-069	H-011	1
T-026	InfraIron	U-070	H-011	1
T-027	PixelPioneers	U-072	H-012	1
T-028	ViewVanguard	U-074	H-013	1
T-029	ScriptSages	U-085	H-013	1
T-030	NodeKnights	U-068	H-015	1
T-031	NexusKnights	U-080	H-013	1
T-032	CloudCommandos	U-081	H-002	1
T-033	SyntaxSquad	U-082	H-007	1
T-034	DevDrafts	U-083	H-022	1
T-035	PixelPioneers	U-084	H-024	1
T-036	CodeCrusaders	U-085	H-014	1

PROJECT:

PK **PK**

team_id	project_title	project_description	github_link
T-001	StatGen	Automated Data Insights	github.com/data-s
T-002	ShieldNet	Network Threat Detector	github.com/sec-s
T-003	HealthTrack	AI Health Monitoring	github.com/byte-h
T-004	CryptoPay	Blockchain Payment App	github.com/chain-p
T-005	SwiftOrder	Food Delivery System	github.com/app-a
T-006	SiteSwift	Fast Web Rendering	github.com/web-w
T-007	SkyStorage	Secure Cloud Vault	github.com/cloud-k
T-008	meshing loops	kuch bhi	NULL
T-009	AI Healthcare Bot	AI-powered diagnostic assistant for rural areas.	https://github.com/404found/ai-health
T-010	BlockChain Vote	Secure electronic voting system using Ethereum.	https://github.com/404found/block-vote
T-011	Cyber Shield	Real-time network intrusion detection system.	https://github.com/404found/cyber-shield
T-012	DataViz Pro	Dynamic data visualization dashboard for sales.	https://github.com/404found/dataviz
T-013	EcoTrack Mobile	Mobile app for tracking personal carbon footprint.	https://github.com/404found/ecotrack
T-014	Smart Grid AI	Optimizing energy distribution using ML.	https://github.com/404found/smartgrid
T-015	Crypto Wallet	Secure multi-currency wallet for web3.	https://github.com/404found/wallet
T-016	Phish Guard	ML model to detect phishing emails and URLs.	https://github.com/404found/phish-guard
T-017	Fleet Manager	Mobile web solution for logistics tracking.	https://github.com/404found/fleet-web
T-018	NFT Market	Decentralized marketplace for digital assets.	https://github.com/404found/nft-hub
T-019	DeepFake Detect	Identifying deepfake videos using CNN.	https://github.com/404found/deep-detect
T-020	Threat Hunter	Cybersecurity tool for vulnerability scanning.	https://github.com/404found/threat-hunt
T-021	MediChain	Patient records management on Blockchain.	https://github.com/404found/medichain
T-022	Arboretum AI	Satellite image analysis for deforestation.	https://github.com/404found/forest-ai
T-023	Zenith Web	E-commerce platform with React and Node.	https://github.com/404found/zenith-shop
T-024	Sentinel Mobile	Mobile security app for Android users.	https://github.com/404found/sentinel
T-025	SupplySafe	Blockchain supply chain transparency tool.	https://github.com/404found/supplysafe
T-026	AutoML Engine	Automated machine learning pipeline for startups.	https://github.com/404found/automl
T-027	ZeroTrust Net	Implementation of zero trust architecture.	https://github.com/404found/zerotrust
T-028	ViewVanguard	Advanced computer vision for security cameras.	https://github.com/404found/viewvanguard
T-029	Aura Blockchain	Privacy-focused transaction protocol.	https://github.com/404found/aura-bc
T-030	Pulse Mobile	Health monitoring app with wearable sync.	https://github.com/404found/pulse-app
T-031	NexusKnights AI	Collaborative AI for swarm robotics.	https://github.com/404found/nexus-ai
T-032	CloudCommandos	Distributed cloud storage with sharding.	https://github.com/404found/cloud-com
T-033	SyntaxSquad Web	PWA for real-time code collaboration.	https://github.com/404found/syntax-squad
T-034	DevDrafts App	Mobile development playground for students.	https://github.com/404found/devdrafts
T-035	PixelPioneers AI	Generative AI for artistic asset creation.	https://github.com/404found/pixel-ai
T-036	CodeCrusaders Cyber	Encrypted messaging service for teams.	https://github.com/404found/code-cyber

TEAM_MEMBER:

PK **PK**

team_id	member_id
T-001	U-068
T-001	U-071
T-001	U-077
T-002	U-069
T-002	U-072
T-002	U-078
T-003	U-070
T-003	U-079
T-003	U-081
T-004	U-070
T-004	U-073
T-004	U-076
T-005	U-074
T-005	U-082
T-005	U-083
T-006	U-075
T-006	U-081
T-006	U-084
T-007	U-079
T-007	U-080
T-007	U-085
T-008	U-082
T-008	U-085
T-008	U-086
T-009	U-068
T-009	U-069
T-010	U-069
T-010	U-070
T-010	U-071
T-011	U-070
T-011	U-072
T-011	U-073
T-012	U-071
T-012	U-074
T-012	U-075
T-013	U-072
T-013	U-076
T-013	U-077
T-014	U-073
T-014	U-078
T-014	U-079
T-015	U-074
T-015	U-080
T-015	U-081
T-016	U-075
T-016	U-082
T-016	U-083
T-017	U-076
T-017	U-084
T-017	U-085
T-018	U-068
T-018	U-070
T-018	U-077
T-019	U-071
T-019	U-072
T-019	U-078
T-020	U-073
T-020	U-074
T-020	U-080
T-021	U-075
T-021	U-076
T-021	U-082
T-022	U-077
T-022	U-078
T-022	U-083
T-023	U-079
T-023	U-080
T-023	U-084
T-024	U-081

T-024	U-082	
T-024	U-085	
T-025	U-069	
T-025	U-083	
T-025	U-084	
T-026	U-069	
T-026	U-070	
T-026	U-085	
T-027	U-068	
T-027	U-072	
T-028	U-074	
T-028	U-076	
T-029	U-078	
T-029	U-080	
T-029	U-085	
T-030	U-068	
T-030	U-082	
T-030	U-084	
T-031	U-069	
T-031	U-071	
T-031	U-080	
T-032	U-073	
T-032	U-075	
T-032	U-081	
T-033	U-077	
T-033	U-079	
T-033	U-082	
T-034	U-081	
T-034	U-083	
T-035	U-070	
T-035	U-084	
T-035	U-085	
T-036	U-072	
T-036	U-074	
T-036	U-085	

HACKATHON:

PK fk FK

hackathon_id	hackathon_name	domain	reg_date	start_date	end_date	prize_pool	admin_id	status_name	judge_id
H-001	Query Quest	Web development	28/05/2026	01/06/2026	03/06/2026	3000	U-001	Upcoming	U-026
H-002	Malware Mania	Web development	22/02/2026	26/02/2026	28/02/2026	4000	U-003	Evaluated	U-026
H-003	Ansible Ace	DevOps	05/06/2026	09/06/2026	11/06/2026	5000	U-021	Upcoming	U-036
H-004	Angular Arena	Web development	25/05/2026	01/06/2026	03/06/2026	4000	U-022	Upcoming	U-026
H-005	DeFi Derby	Blockchain	20/05/2026	24/05/2026	26/05/2026	5000	U-024	Upcoming	U-038
H-006	ChatBot Championship	AI & ML	01/03/2026	05/03/2026	07/03/2026	3000	U-025	Evaluated	U-030
H-007	Neural Network Ninja	AI & ML	10/04/2026	14/04/2026	16/04/2026	5000	U-002	Evaluated	U-029
H-008	NFT Knights	Blockchain	27/04/2026	01/05/2026	03/05/2026	4000	U-004	Ongoing	U-039
H-009	Jenkins Jive	DevOps	27/04/2026	01/05/2026	03/05/2026	4500	U-007	Ongoing	U-040
H-010	Chain Reaction	Blockchain	05/04/2026	09/04/2026	11/04/2026	6000	U-009	Evaluated	U-034
H-011	React Race	Web Development	27/04/2026	01/05/2026	03/05/2026	3000	U-010	Ongoing	U-041
H-012	Predictive Pros	AI & ML	27/04/2026	01/05/2026	03/05/2026	5500	U-013	Ongoing	U-042
H-013	Full Stack Fever	Web Development	27/04/2026	01/05/2026	03/05/2026	5000	U-014	Ongoing	U-027
H-014	Cloud Castles	Cloud Computing	02/04/2026	06/04/2026	08/04/2026	5000	U-015	Evaluated	U-028
H-015	Pipeline Pro	DevOps	27/04/2026	01/05/2026	03/05/2026	6500	U-016	Ongoing	U-036
H-016	Pandas Play	Data Science	20/04/2026	24/04/2026	26/04/2026	4000	U-018	Evaluated	U-032
H-017	Swift Safari	Mobile App Dev	22/04/2026	26/04/2026	28/04/2026	5000	U-019	Evaluated	U-034
H-018	Shield Sprint	Cybersecurity	12/04/2026	16/04/2026	18/04/2026	5000	U-020	Evaluated	U-026
H-019	Serverless Summit	Cloud Computing	20/03/2026	24/03/2026	26/03/2026	4500	U-005	Evaluated	U-035
H-020	Deep Dive Vision	AI & ML	15/06/2026	19/06/2026	21/06/2026	4500	U-006	Upcoming	U-037
H-021	Git Guide	DevOps	18/09/2026	22/09/2026	24/09/2026	3000	U-008	Upcoming	U-038
H-022	Solana Sprint	Blockchain	10/04/2026	14/04/2026	16/04/2026	7000	U-011	Evaluated	U-043
H-023	Pentest Party	Cybersecurity	01/10/2026	05/10/2026	07/10/2026	6500	U-012	Upcoming	U-039
H-024	Flutter Flow	Mobile App Dev	10/02/2026	14/02/2026	16/02/2026	4500	U-017	Evaluated	U-031
H-025	NLP Nexus	AI & ML	12/08/2026	16/08/2026	18/08/2026	4000	U-023	Upcoming	U-040

WORKSHOP:

PK **FK** **FK** **FK**

workshop_id	workshop_name	workshop_url	workshop_date	start_time	end_time	admin_id	mentor_id	hackathon_id
W-001	Advanced SQL Queries	https://sql-pro.io/adv	28/04/2026	11:00:00	13:00:00	U-001	U-047	H-013
W-002	Intro to Neural Networks	https://learn-ai.io/nn-101	11/04/2026	10:00:00	12:00:00	U-002	U-056	H-007
W-003	Malware Analysis	https://mal-detect.com/lab	23/02/2026	15:00:00	18:00:00	U-003	U-045	H-002
W-004	NFT Minting Guide	https://nft-school.xyz/mint	28/04/2026	16:00:00	18:00:00	U-004	U-044	H-008
W-005	Serverless Functions	https://lambda-labs.xyz/serv	21/03/2026	9:00:00	11:00:00	U-005	U-059	H-019
W-006	Computer Vision Basics	https://vision-academy.com/cv	16/06/2026	14:00:00	16:00:00	U-006	U-054	H-020
W-007	CI/CD with Jenkins	https://jenkins-guide.io/cicd	28/04/2026	10:00:00	12:00:00	U-007	U-065	H-009
W-008	Git & GitHub Mastery	https://git-pro.com/workflow	19/09/2026	14:00:00	16:00:00	U-008	U-050	H-021
W-009	Solidity Smart Contracts	https://eth-build.io/solidity	21/05/2026	13:00:00	15:00:00	U-009	U-057	H-005
W-010	Modern React.js	https://react-dev.com/hooks	28/04/2026	10:00:00	12:00:00	U-010	U-061	H-011
W-011	Building on Solana	https://solana-dev.net/workshop	11/04/2026	14:00:00	17:00:00	U-011	U-062	H-022
W-012	Network Pentesting	https://cyber-ops.io/pentest	13/04/2026	14:00:00	17:00:00	U-012	U-051	H-018
W-013	PyTorch for Beginners	https://pytorch-labs.edu/intro	21/04/2026	9:00:00	11:00:00	U-013	U-058	H-016
W-014	Tailwind CSS	https://style-pro.xyz/tailwind	26/05/2026	9:00:00	11:00:00	U-014	U-051	H-004
W-015	AWS Essentials	https://aws-cloud.com/basics	03/04/2026	10:00:00	12:00:00	U-015	U-052	H-014
W-016	Site Reliability (SRE)	https://sre-pros.com/basics	28/04/2026	15:00:00	17:00:00	U-016	U-046	H-015
W-017	Flutter Framework	https://flutter-dev.io/intro	11/02/2026	10:00:00	13:00:00	U-017	U-055	H-024
W-018	Python for Data Science	https://py-data.org/course	29/05/2026	10:00:00	13:00:00	U-018	U-060	H-001
W-019	Swift UI for iOS	https://ios-academy.com/swift	23/04/2026	14:00:00	16:00:00	U-019	U-066	H-017
W-020	Ethical Hacking 101	https://hack-shield.com/eh	02/10/2026	10:00:00	13:00:00	U-020	U-049	H-023
W-021	Ansible Automation	https://ansible-labs.xyz/intro	07/02/2026	11:00:00	13:00:00	U-021	U-063	H-003
W-022	Mastering MongoDB	https://db-school.com/mongo	29/05/2026	11:00:00	13:00:00	U-022	U-048	H-001
W-023	Ethics in AI	https://ethics-tech.com/workshop	02/03/2026	15:00:00	16:30:00	U-023	U-067	H-006
W-024	Hyperledger Fabric	https://blockchain-pro.com/hl	06/04/2026	11:00:00	13:00:00	U-024	U-064	H-010
W-025	Natural Language Processing	https://nlp-hub.org/start	13/08/2026	11:00:00	13:00:00	U-025	U-053	H-025

WORKSHOP_ATTENDEES:

PK **PK**

participant_id	workshop_id
U-068	W-001
U-070	W-001
U-075	W-001
U-078	W-001
U-068	W-002
U-069	W-002
U-084	W-002
U-068	W-003
U-070	W-003
U-072	W-003
U-074	W-003
U-075	W-003
U-082	W-003
U-085	W-003
U-068	W-004
U-073	W-004
U-076	W-004
U-070	W-005
U-080	W-005
U-070	W-006
U-078	W-007
U-083	W-007
U-079	W-008
U-069	W-010
U-070	W-010
U-074	W-010
U-076	W-010
U-078	W-010

U-085	W-024
U-079	W-025
U-081	W-025
U-085	W-025

SPONSOR:

PK **FK**

sponsor_id	sponsor_name	status	admin_id	website_url	email
S-001	Apple	Accepted	U-001	https://www.apple.com	contact@apple.com
S-002	AWS	Accepted	U-001	https://aws.amazon.com	info@aws.amazon.com
S-003	Azure	Accepted	U-001	https://azure.microsoft.com	partner@microsoft.com
S-004	Binance	Accepted	U-002	https://www.binance.com	support@binance.com
S-005	CircleCI	Accepted	U-002	https://circleci.com	hello@circleci.com
S-006	CloudBees	Accepted	U-002	https://www.cloudbees.com	info@cloudbees.com
S-007	CrowdStrike	Accepted	U-003	https://www.crowdstrike.com	security@crowdstrike.com
S-008	Databricks	Accepted	U-003	https://www.databricks.com	data@databricks.com
S-009	Ethereum Foundation	Accepted	U-003	https://ethereum.org	info@ethereum.org
S-010	FireEye	Accepted	U-004	https://www.fireeye.com	threats@fireeye.com
S-011	GitHub	Accepted	U-004	https://github.com	partnership@github.com
S-012	Google	Accepted	U-004	https://www.google.com	ads@google.com
S-013	HuggingFace	Accepted	U-005	https://huggingface.co	model@huggingface.co
S-014	Meta	Pending	U-005	https://about.meta.com	business@meta.com
S-015	NVIDIA	Pending	U-006	https://www.nvidia.com	info@nvidia.com
S-016	OpenAI	Pending	U-006	https://openai.com	api@openai.com
S-017	OpenSea	Pending	U-006	https://opensea.io	nft@opensea.io
S-018	Palo Alto Networks	Accepted	U-007	https://www.paloaltonetworks.com	sales@paloaltonetworks.com
S-019	Postman	Accepted	U-007	https://www.postman.com	api@postman.com
S-020	RedHat	Accepted	U-007	https://www.redhat.com	support@redhat.com
S-021	Snowflake	Pending	U-008	https://www.snowflake.com	cloud@snowflake.com
S-022	Solana Labs	Rejected	U-001	https://solana.com	dev@solana.com
S-023	Toyota	Accepted	U-001	https://www.toyota.com	corporate@toyota.com
S-024	Vercel	Accepted	U-009	https://vercel.com	contact@vercel.com
S-025	AWS	Accepted	U-001	Funding for Ansible Ace	REQ_10000.0
S-026	IZMA	Accepted	U-001	org-bilal12@gmail.com	https://www.herefg.com
S-027	AWS	Accepted	U-001	Funding for Query Quest	REQ_6000.0

SPONSOR_HACKATHON:

PK **PK**

sponsor_id	hackathon_id	funding_amount
S-001	H-001	6000
S-001	H-010	8000
S-001	H-024	5500
S-002	H-001	4000
S-002	H-010	8000
S-002	H-016	5000
S-003	H-001	4000
S-003	H-011	4500
S-003	H-016	5000
S-004	H-007	6500
S-004	H-011	4500
S-004	H-016	5000
S-005	H-007	6500
S-005	H-011	4500
S-005	H-017	7000
S-006	H-007	6500
S-006	H-017	7000
S-006	H-022	9000
S-007	H-002	5000
S-007	H-017	7000
S-007	H-022	9000
S-008	H-002	5000
S-008	H-018	7000
S-008	H-022	9000
S-009	H-002	5000
S-009	H-018	7000
S-009	H-023	8000
S-010	H-008	5500

S-010	H-018	7000
S-010	H-023	8000
S-011	H-003	6500
S-011	H-008	5500
S-011	H-023	8000
S-012	H-001	7000
S-012	H-003	6500
S-012	H-008	5500
S-012	H-012	7000
S-012	H-019	5500
S-013	H-004	5000
S-013	H-012	7000
S-013	H-019	5500
S-014	H-004	5000
S-014	H-013	6500
S-014	H-019	5500
S-015	H-004	5000
S-015	H-013	6500
S-015	H-020	5500
S-016	H-013	6500
S-016	H-020	5500
S-016	H-025	5000
S-017	H-014	6500
S-017	H-020	5500
S-017	H-025	5000
S-018	H-009	5500
S-018	H-014	6500
S-018	H-025	5000
S-019	H-005	6000
S-019	H-009	5500
S-019	H-014	6500
S-020	H-005	6000
S-020	H-009	5500
S-020	H-015	8000
S-021	H-005	6000
S-021	H-015	8000
S-021	H-021	4000
S-022	H-006	4000
S-022	H-015	8000
S-022	H-021	4000
S-023	H-006	4000
S-023	H-021	4000
S-023	H-024	5500
S-024	H-006	4000
S-024	H-010	8000
S-024	H-024	5500

SCORE:

PK **FK** **PK** **PK**

judge_id	hackathon_id	team_id	project_title	technical_score	evaluative_score	innovative_score	Final_Score
U-026	H-018	T-002	ShieldNet	8	7	10	25
U-028	H-014	T-036	CodeCrusaders Cyber	8	7	5	20
U-029	H-007	T-033	SyntaxSquad Web	6	10	9	25
U-030	H-006	T-003	HealthTrack	6	7	7	20
U-031	H-024	T-035	PixelPioneers AI	9	8	9	26
U-032	H-016	T-001	StatGen	7	10	8	25
U-033	H-002	T-032	CloudCommandos	5	6	7	18
U-034	H-010	T-004	CryptoPay	8	10	7	25
U-034	H-017	T-005	SwiftOrder	9	10	10	29
U-035	H-019	T-007	SkyStorage	9	9	10	28
U-039	H-008	T-021	MediChain	7	8	6	21
U-039	H-008	T-023	Zenith Web	9	8	9	26
U-043	H-022	T-034	DevDrafts App	9	8	8	25

11.3 Third Normal Form (3NF):

It requires the table to be in 2NF and ensures that no non-key attribute depends on another non-key attribute, effectively removing all transitive dependencies.

Transitive Dependency

Hackathon_ID → (Commencement_date, End_date) → Hackathon_status

(Technical_score, Innovative_score, Executive_score) → Final_score

HACKATHON:

PK **FK** **fk** **fk**

hackathon_id	hackathon_name	domain	reg_date	start_date	end_date	prize_pool	admin_id	status_id	judge_id
H-001	Query Quest	Web development	28/05/2026	01/06/2026	03/06/2026	3000	U-001	1	U-026
H-002	Malware Mania	Web development	22/02/2026	26/02/2026	28/02/2026	4000	U-003	2	U-026
H-003	Ansible Ace	DevOps	05/06/2026	09/06/2026	11/06/2026	5000	U-021	3	U-036
H-004	Angular Arena	Web development	25/05/2026	01/06/2026	03/06/2026	4000	U-022	4	U-026
H-005	DeFi Derby	Blockchain	20/05/2026	24/05/2026	26/05/2026	5000	U-024	5	U-038
H-006	ChatBot Championship	AI & ML	01/03/2026	05/03/2026	07/03/2026	3000	U-025	6	U-030
H-007	Neural Network Ninja	AI & ML	10/04/2026	14/04/2026	16/04/2026	5000	U-002	7	U-029
H-008	NFT Knights	Blockchain	27/04/2026	01/05/2026	03/05/2026	4000	U-004	8	U-039
H-009	Jenkins Jive	DevOps	27/04/2026	01/05/2026	03/05/2026	4500	U-007	9	U-040
H-010	Chain Reaction	Blockchain	05/04/2026	09/04/2026	11/04/2026	6000	U-009	10	U-034
H-011	React Race	Web Development	27/04/2026	01/05/2026	03/05/2026	3000	U-010	11	U-041
H-012	Predictive Pros	AI & ML	27/04/2026	01/05/2026	03/05/2026	5500	U-013	12	U-042
H-013	Full Stack Fever	Web Development	27/04/2026	01/05/2026	03/05/2026	5000	U-014	13	U-027
H-014	Cloud Castles	Cloud Computing	02/04/2026	06/04/2026	08/04/2026	5000	U-015	14	U-028
H-015	Pipeline Pro	DevOps	27/04/2026	01/05/2026	03/05/2026	6500	U-016	15	U-036
H-016	Pandas Play	Data Science	20/04/2026	24/04/2026	26/04/2026	4000	U-018	16	U-032
H-017	Swift Safari	Mobile App Dev	22/04/2026	26/04/2026	28/04/2026	5000	U-019	17	U-034
H-018	Shield Sprint	Cybersecurity	12/04/2026	16/04/2026	18/04/2026	5000	U-020	18	U-026
H-019	Serverless Summit	Cloud Computing	20/03/2026	24/03/2026	26/03/2026	4500	U-005	19	U-035
H-020	Deep Dive Vision	AI & ML	15/06/2026	19/06/2026	21/06/2026	4500	U-006	20	U-037
H-021	Git Guide	DevOps	18/09/2026	22/09/2026	24/09/2026	3000	U-008	21	U-038
H-022	Solana Sprint	Blockchain	10/04/2026	14/04/2026	16/04/2026	7000	U-011	22	U-043
H-023	Pentest Party	Cybersecurity	01/10/2026	05/10/2026	07/10/2026	6500	U-012	23	U-039
H-024	Flutter Flow	Mobile App Dev	10/02/2026	14/02/2026	16/02/2026	4500	U-017	24	U-031
H-025	NLP Nexus	AI & ML	12/08/2026	16/08/2026	18/08/2026	4000	U-023	25	U-040

STATUS_MASTER:

PK

status_id	status_name
1	Upcoming
2	Evaluated
3	Upcoming
4	Upcoming
5	Upcoming
6	Evaluated
7	Evaluated
8	Ongoing
9	Ongoing
10	Evaluated
11	Ongoing
12	Ongoing
13	Ongoing
14	Evaluated
15	Ongoing
16	Evaluated
17	Evaluated
18	Evaluated
19	Evaluated
20	Upcoming
21	Upcoming
22	Evaluated
23	Upcoming
24	Evaluated
25	Upcoming

SCORE:

PK FK PK PK

judge_id	hackathon_id	team_id	project_title	technical_score	evaluative_score	innovative_score
U-026	H-018	T-002	ShieldNet	8	7	10
U-028	H-014	T-036	CodeCrusaders Cyber	8	7	5
U-029	H-007	T-033	SyntaxSquad Web	6	10	9
U-030	H-006	T-003	HealthTrack	6	7	7
U-031	H-024	T-035	PixelPioneers AI	9	8	9
U-032	H-016	T-001	StatGen	7	10	8
U-033	H-002	T-032	CloudCommandos	5	6	7
U-034	H-010	T-004	CryptoPay	8	10	7
U-034	H-017	T-005	SwiftOrder	9	10	10
U-035	H-019	T-007	SkyStorage	9	9	10
U-039	H-008	T-021	MediChain	7	8	6
U-039	H-008	T-023	Zenith Web	9	8	9
U-043	H-022	T-034	DevDrafts App	9	8	8

To achieve Third Normal Form (3NF), we removed the Final_sum attribute from the physical database schema. This was necessary because Final_sum created a transitive dependency, as its value depended on other non-key attributes (Technical_score, Innovative_score, and Executive_score) rather than the Primary Key alone.

Instead of moving this to a separate table, we have implemented the summation logic at the UI/Application level. By calculating the total score dynamically during data display, we eliminate data redundancy and ensure that the final result is always synchronized with the individual scoring metrics.

11.4 Boyce-Codd Normal Form (BCNF):

For every functional dependency X → Y, X must be a superkey (candidate key).

Table	Functional Dependency	X is Superkey?	BCNF?
`User_Master`	*user_id → all attributes*	**Yes (PK)**	✓ Yes
`Team`	*team_id → all attributes*	**Yes (PK)**	✓ Yes
`Project`	*(team_id, project_title) → all attributes*	**Yes (Composite PK)**	✓ Yes
`Team_Members`	*(team_id, member_id) → all attributes*	**Yes (Composite PK)**	✓ Yes
`Hackathon`	*hackathon_id → all attributes*	**Yes (PK)**	✓ Yes
`Workshop`	*workshop_id → all attributes*	**Yes (PK)**	✓ Yes
`Workshop_Attendees`	*(participant_id, workshop_id) → all attributes*	**Yes (Composite PK)**	✓ Yes
`Sponsor`	*sponsor_id → all attributes*	**Yes (PK)**	✓ Yes
`Sponsor_Hackathon`	*(sponsor_id, hackathon_id) → all attributes*	**Yes (Composite PK)**	✓ Yes
`Scores (Junction Table)`	*(judge_id, team_id, project_title) → all attributes*	**Yes (Composite PK)**	✓ Yes
`Status_Master`	*status_id → all attributes*	**Yes (PK)**	✓ Yes

Every determinant in every table is a superkey.

12. Operational Governance

System Rules (Operational Governance)

These hardcoded constraints are the "Laws of the System" that maintain technical and competitive integrity. Any future developer or auditor should note these mandatory enforcements:

Rule Name	Description
Team Composition Limit	The system strictly caps team registration at a maximum of 3 members per team. This is a hard constraint to ensure fair competition.
Proposal-Based Registration	A team's status is marked as 'Pending' upon initial signup; full registration is only finalized once the Admin approves the Project Idea/Proposal.
Identity Singularity	The User_Email attribute is defined as a Unique Key; duplicate accounts or multi-role registrations under a single email are strictly prohibited.
Evaluation Pre-requisites	The scoring module is programmatically locked for any team that has not received Admin approval or has failed to submit a valid GitHub repository link.
Atomic Storage Policy	Following 3NF standards, the database forbids the storage of derived values (e.g., Total Scores). All calculations are performed on-the-fly via SQL Views to prevent data staleness.
Technical Boundary	To maintain backend performance and reduce architectural complexity, the system exclusively supports SQL-based relational data and restricts the use of external NoSQL structures or complex multimedia URLs.
Mandatory Lead Assignment	Every team must designate exactly one Team Lead who holds the exclusive authority to edit project descriptions and submit final links.

13 Technology Stack

The project leverages a modern, Python-centric stack for efficiency and performance:

Technology	Role in Project
Streamlit	Used as the primary framework for building the interactive UI and role-based dashboard navigation.
Python	The core engine for business logic, backend processing, and integration.
MySQL	The relational database management system used for physical data modeling and 3NF-compliant storage.
Pandas	Utilized for high-speed data manipulation, cleaning messy datasets, and preparing data for database insertion.
Plotly	Employed to generate visual analytics and performance charts for the Admin and Judge dashboards.
HTML/CSS	Custom styling injected into Streamlit to enhance the aesthetic and professional feel of the user interfaces.

5. Physical Model

1. STATUS_MASTER:

```
CREATE TABLE status_master (

    status_id INT PRIMARY KEY,

    status_name VARCHAR(50) NOT NULL

);

INSERT INTO status_master (status_id, status_name)

SELECT * FROM hackathon_db.hack_status;
```

2. USER_MASTER:

```
CREATE TABLE user_master (

    user_id VARCHAR(50) PRIMARY KEY,

    user_name VARCHAR(100) NOT NULL,

    email VARCHAR(225) UNIQUE,

    phone_no VARCHAR(20),

    role VARCHAR(50)

);

INSERT INTO user_master (email, user_name, phone_no, role, user_id)

SELECT User_email, User_name, User_phone, User_role, User_id

FROM hackathon_db.user_table;
```

3. HACKATHON_TABLE:

```
CREATE TABLE hackathon_table (

    hackathon_id VARCHAR(50) PRIMARY KEY,

    hackathon_name VARCHAR(255) NOT NULL,

    domain VARCHAR(100),

    reg_date DATE,
```

```
    start_date DATE,

    end_date DATE,

    prize_pool DECIMAL(15, 2),

    admin_id VARCHAR(50),

    status_id INT,

    FOREIGN KEY (admin_id) REFERENCES user_master(user_id),

    FOREIGN KEY (status_id) REFERENCES status_master(status_id)

);

INSERT INTO hackathon_table (hackathon_id, hackathon_name,admin_id,
domain, reg_date, start_date, end_date, prize_pool, status_id)

SELECT

Hackathon_id,

Hackathon_title,

Admin_id,

Hackathon_domain,

STR_TO_DATE(`Reg-deadline`, '%m/%d/%Y'),

STR_TO_DATE(Commencement_date, '%m/%d/%Y'),

STR_TO_DATE(End_date, '%m/%d/%Y'),

Prize_pool,

Status_id

FROM hackathon_db.hack_table;
```

4. WORKSHOP_TABLE:

```
CREATE TABLE workshop_table (

    workshop_id VARCHAR(50) PRIMARY KEY,

    workshop_name VARCHAR(255) NOT NULL,
```

```sql
    workshop_url VARCHAR(500),

    workshop_date DATE,

    start_time TIME,

    end_time TIME,

    admin_id VARCHAR(50),

    mentor_id VARCHAR(50),

    hackathon_id VARCHAR(50),

    FOREIGN KEY (admin_id) REFERENCES user_master(user_id),

    FOREIGN KEY (mentor_id) REFERENCES user_master(user_id),

    FOREIGN KEY (hackathon_id) REFERENCES hackathon_table(hackathon_id)

);

INSERT INTO workshop_table (workshop_id, workshop_name, workshop_url,
workshop_date, start_time, end_time, admin_id, mentor_id, hackathon_id)


SELECT

    Workshop_id,

    Workshop_title,

    Workshop_URL,

    STR_TO_DATE(Date, '%m/%d/%Y'),

    STR_TO_DATE(Workshop_start_time, '%h:%i:%s %p'),

    STR_TO_DATE(Workshop_end_time, '%h:%i:%s %p'),

    Admin_id,

    Mentor_id,

    Hackathon_id

FROM hackathon_db.workshop_hack;
```

5. **SPONSOR_TABLE:**

```sql
CREATE TABLE sponsor_table (

    sponsor_id VARCHAR(50) PRIMARY KEY,

    sponsor_name VARCHAR(255) NOT NULL,

    status VARCHAR(50),

    admin_id VARCHAR(50),

    FOREIGN KEY (admin_id) REFERENCES user_master(user_id)

);

INSERT INTO sponsor_table (sponsor_id, sponsor_name, status, admin_id)

SELECT

    Sponsor_id,

    Sponsor_name,

    Sponsor_status,

    Admin_id

FROM hackathon_db.sponsor_table;
```

6. **SPONSOR_HACKATHON:**

```sql
CREATE TABLE sponsor_hackathon (

    sponsor_id VARCHAR(50),

    hackathon_id VARCHAR(50),

    funding_amount DECIMAL(15, 2),

    PRIMARY KEY (sponsor_id, hackathon_id),

    FOREIGN KEY (sponsor_id) REFERENCES sponsor_table(sponsor_id),

    FOREIGN KEY (hackathon_id) REFERENCES hackathon_table(hackathon_id)

);
```

```sql
INSERT INTO sponsor_hackathon (sponsor_id, hackathon_id, funding_amount)

SELECT

    Sponsor_id,

    Hackathon_id,

    Sponsor_amount

FROM hackathon_db.sponsor_hack_table;
```

7. TEAM:

```sql
CREATE TABLE team (

    team_id VARCHAR(50) PRIMARY KEY,

    team_name VARCHAR(255) NOT NULL,

    lead_id VARCHAR(50),

    hackathon_id VARCHAR(50),

    registration_status BOOLEAN DEFAULT FALSE,

    FOREIGN KEY (lead_id) REFERENCES user_master(user_id),

    FOREIGN KEY (hackathon_id) REFERENCES hackathon_table(hackathon_id)

);

INSERT INTO team (team_id, team_name, lead_id, hackathon_id,
registration_status)

SELECT

    Team_id,

    Team_name,

    Lead_id,

    Hackathon_id,

    CASE

        WHEN Registration_status = 'TRUE' THEN 1

        WHEN Registration_status = 'FALSE' THEN 0
```

```
        ELSE 0

    END

FROM hackathon_db.teams_table;
```

8. TEAM_MEMBERS:

```
CREATE TABLE team_members (

    team_id VARCHAR(50),

    member_id VARCHAR(50),

    PRIMARY KEY (team_id, member_id),

    FOREIGN KEY (team_id) REFERENCES team(team_id),

    FOREIGN KEY (member_id) REFERENCES user_master(user_id)

);

INSERT INTO team_members (team_id, member_id)

SELECT

    Team_id,

    Member_id

FROM hackathon_db.team_members_table;
```

9. PROJECT:

```
CREATE TABLE project (

    team_id VARCHAR(50),

    project_title VARCHAR(255),

    project_description TEXT,

    github_link VARCHAR(500),

    PRIMARY KEY (team_id, project_title),

    FOREIGN KEY (team_id) REFERENCES team(team_id)
```

```
);

TRUNCATE TABLE project;

INSERT INTO project (team_id, project_title, project_description,
github_link)

SELECT

    Team_id,

    Project_name,

    Project_description,

    `Github-link`

FROM hackathon_db.project_table;
```

10. SCORES:

```
CREATE TABLE scores (

    judge_id VARCHAR(50),

    hackathon_id VARCHAR(50),

    team_id VARCHAR(50),

    project_title VARCHAR(255),

    technical_score INT CHECK (technical_score <= 10),user_master

    evaluative_score INT CHECK (evaluative_score <= 10),

    innovative_score INT CHECK (innovative_score <= 10),

    PRIMARY KEY (judge_id, team_id, project_title),

    FOREIGN KEY (judge_id) REFERENCES user_master(user_id),

    FOREIGN KEY (hackathon_id) REFERENCES hackathon_table(hackathon_id),

    FOREIGN KEY (team_id, project_title) REFERENCES project(team_id,
project_title)

);

INSERT INTO scores (judge_id, hackathon_id, team_id, project_title,
technical_score, evaluative_score, innovative_score)
```

```sql
SELECT

    Judge_id,

    Hackathon_id,

    Team_id,

    Project_name,

    Technical_score,

    Executive_score,

    Innovative_score

FROM hackathon_db.scoresss;
```

11. WORKSHOP_ATTENDEES:

```sql
CREATE TABLE workshop_attendees (

    participant_id VARCHAR(50),

    workshop_id VARCHAR(50),

    PRIMARY KEY (participant_id, workshop_id),

    FOREIGN KEY (participant_id) REFERENCES user_master(user_id),

    FOREIGN KEY (workshop_id) REFERENCES workshop_table(workshop_id)

);

INSERT INTO workshop_attendees (participant_id, workshop_id)

SELECT

    Participant_id,

    Workshop_id

FROM hackathon_db.workshop_attendees;
```

Specifying Constraints:

```
ALTER TABLE hackathon_table

DROP FOREIGN KEY IF EXISTS hackathon_table_ibfk_1,

DROP FOREIGN KEY IF EXISTS hackathon_table_ibfk_2,

DROP FOREIGN KEY IF EXISTS fk_hackathon_judge;



ALTER TABLE hackathon_table

ADD CONSTRAINT fk_hack_admin FOREIGN KEY (admin_id) REFERENCES
user_master(user_id)

    ON UPDATE CASCADE ON DELETE RESTRICT,

ADD CONSTRAINT fk_hack_status FOREIGN KEY (status_id) REFERENCES
status_master(status_id)

    ON UPDATE CASCADE ON DELETE CASCADE,

ADD CONSTRAINT fk_hack_judge FOREIGN KEY (judge_id) REFERENCES
user_master(user_id)

    ON UPDATE CASCADE ON DELETE SET NULL;

ALTER TABLE workshop_table

ADD CONSTRAINT fk_work_admin FOREIGN KEY (admin_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE RESTRICT,

ADD CONSTRAINT fk_work_mentor FOREIGN KEY (mentor_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE SET NULL,

ADD CONSTRAINT fk_work_hack FOREIGN KEY (hackathon_id) REFERENCES
hackathon_table(hackathon_id) ON UPDATE CASCADE ON DELETE CASCADE;
```

```
ALTER TABLE sponsor_table
```

```sql
ADD CONSTRAINT fk_spon_admin FOREIGN KEY (admin_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE RESTRICT;

ALTER TABLE sponsor_hackathon
ADD CONSTRAINT fk_sh_sponsor FOREIGN KEY (sponsor_id) REFERENCES
sponsor_table(sponsor_id) ON UPDATE CASCADE ON DELETE CASCADE,
ADD CONSTRAINT fk_sh_hack FOREIGN KEY (hackathon_id) REFERENCES
hackathon_table(hackathon_id) ON UPDATE CASCADE ON DELETE CASCADE;
ALTER TABLE team
ADD CONSTRAINT fk_team_lead FOREIGN KEY (lead_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE RESTRICT,
ADD CONSTRAINT fk_team_hack FOREIGN KEY (hackathon_id) REFERENCES
hackathon_table(hackathon_id) ON UPDATE CASCADE ON DELETE CASCADE;
ALTER TABLE team_members
ADD CONSTRAINT fk_tm_team FOREIGN KEY (team_id) REFERENCES team(team_id) ON
UPDATE CASCADE ON DELETE CASCADE,
ADD CONSTRAINT fk_tm_member FOREIGN KEY (member_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE CASCADE;
ALTER TABLE project
ADD CONSTRAINT fk_project_team FOREIGN KEY (team_id) REFERENCES team(team_id)
ON UPDATE CASCADE ON DELETE CASCADE;
```

```sql
ALTER TABLE scores

ADD CONSTRAINT fk_score_judge FOREIGN KEY (judge_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE RESTRICT,

ADD CONSTRAINT fk_score_hack FOREIGN KEY (hackathon_id) REFERENCES
hackathon_table(hackathon_id) ON UPDATE CASCADE ON DELETE CASCADE,

ADD CONSTRAINT fk_score_project FOREIGN KEY (team_id, project_title) REFERENCES
project(team_id, project_title) ON UPDATE CASCADE ON DELETE CASCADE;

TABLE scores

ADD CONSTRAINT fk_score_judge FOREIGN KEY (judge_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE RESTRICT,

ADD CONSTRAINT fk_score_hack FOREIGN KEY (hackathon_id) REFERENCES
hackathon_table(hackathon_id) ON UPDATE CASCADE ON DELETE CASCADE,

ADD CONSTRAINT fk_score_project FOREIGN KEY (team_id, project_title) REFERENCES
project(team_id, project_title) ON UPDATE CASCADE ON DELETE CASCADE;
ALTER TABLE workshop_attendees
ADD CONSTRAINT fk_wa_participant FOREIGN KEY (participant_id) REFERENCES
user_master(user_id) ON UPDATE CASCADE ON DELETE CASCADE,
ADD CONSTRAINT fk_wa_workshop FOREIGN KEY (workshop_id) REFERENCES
workshop_table(workshop_id) ON UPDATE CASCADE ON DELETE CASCADE;
```

Alter tables

Due to business requirements for adding further features we alter tables as follows :

ALTER USER MASTER BY ADDING PASSWORD COLUMN FOR SECURITY

1:

```sql
ALTER TABLE user_master
ADD COLUMN password VARCHAR(255) DEFAULT '12345';

SET SQL_SAFE_UPDATES = 0;

UPDATE user_master
SET password = 'admin786'
WHERE role = 'Admin Staff';
```

```sql
UPDATE user_master
SET password = 'judge2026'
WHERE role = 'Judge';


UPDATE user_master
SET password = 'mentor123'
WHERE role = 'Mentor';

SET SQL_SAFE_UPDATES = 1;
```

user_id	user_name	email	phone_no	role	password
U-001	Ahad Raza Mir	org-ahad20@gmail.com	0334-5556667	Admin Staff	admin786
U-002	Ahmed Raza	org-ahmed1@gmail.com	0300-1112223	Admin Staff	admin786
U-003	Bilal Saeed	org-bilal12@gmail.com	0328-2223334	Admin Staff	admin786
U-004	Bushra Ansari	org-bush9@gmail.com	0332-4445556	Admin Staff	admin786
U-005	Fahad Mustafa	org-fahad14@gmail.com	0340-8889990	Admin Staff	admin786
U-006	Fatima Sohail	org-fatima2@gmail.com	0321-4445556	Admin Staff	admin786
U-007	Hamza Ali	org-hamza18@gmail.com	0313-0001112	Admin Staff	admin786
U-008	Hania Aamir	org-hania23@gmail.com	0314-4445556	Admin Staff	admin786
U-009	Hassan Ali	org-hassan6@gmail.com	0305-5556667	Admin Staff	admin786
U-010	Imran Abbas	org-imran16@gmail.com	0329-4445556	Admin Staff	admin786
U-011	Iqra Aziz	org-iqra7@gmail.com	0318-8889990	Admin Staff	admin786
U-012	Kubra Khan	org-kub11@gmail.com	0309-0001112	Admin Staff	admin786
U-013	Marium Khan	org-mari4@gmail.com	0345-0001112	Admin Staff	admin786
U-014	Maya Ali	org-maya21@gmail.com	0342-7778889	Admin Staff	admin786
U-015	Mehwish Hayat	org-meh13@gmail.com	0339-5556667	Admin Staff	admin786
U-016	Momina Mustehsan	org-mom25@gmail.com	0335-0001112	Admin Staff	admin786
U-017	Osman Khalid	org-osman22@gmail.com	0303-1112223	Admin Staff	admin786
U-018	Sajal Ali	org-sajal15@gmail.com	0311-1112223	Admin Staff	admin786
U-019	Sanam Baloch	org-sanam17@gmail.com	0302-7778889	Admin Staff	admin786

adding probem statemnt to make hackthon purpose undestandable to users

2:

```sql
ALTER TABLE hackathon_table
ADD COLUMN problem_statement TEXT;



SET SQL_SAFE_UPDATES = 0;
UPDATE hackathon_table
SET problem_statement = CASE
    WHEN domain = 'AI & ML' THEN 'Develop a predictive model or generative AI solution to solve complex real-world data challenges.'
    WHEN domain = 'Blockchain' THEN 'Build a decentralized application (dApp) focusing on transparency, security, and smart contract automation.'
    WHEN domain = 'Cloud Computing' THEN 'Design a scalable, serverless architecture to optimize resource management and high availability.'
    WHEN domain = 'Cybersecurity' THEN 'Create a tool for real-time threat detection, data encryption, or vulnerability assessment in networked systems.'
    WHEN domain = 'Data Science' THEN 'Perform deep exploratory data analysis to derive actionable insights and visualize trends from large datasets.'
    WHEN domain = 'DevOps' THEN 'Automate CI/CD pipelines and infrastructure monitoring to enhance deployment speed and system reliability.'
```

```sql
    WHEN domain = 'Mobile App Dev' THEN 'Create a user-centric mobile application with cross-platform compatibility and seamless offline functionality.'
    WHEN domain = 'Web Development' THEN 'Develop a high-performance, responsive web application using modern frameworks with secure backend integration.'
    ELSE 'Create an innovative technical solution tailored to the specific needs of this domain.'
END;
SET SQL_SAFE_UPDATES = 1;
ALTER TABLE team ADD COLUMN admin_approval VARCHAR(20) DEFAULT 'Pending';

SET SQL_SAFE_UPDATES = 0;

-- Team 1: Cybersecurity
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-001';

-- Team 2: Data Science
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-002';

-- Team 3: Blockchain
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-003';

-- Team 4: AI/ML
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-004';

-- Team 5: DevOps (Automation focus)
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-005';
```

```sql
-- Team 6: DevOps (Monitoring focus)
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-006';

-- Team 7: Cybersecurity (Network focus)
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-007';
```

hackathon_id	hackathon_name	domain	reg_date	start_date	end_date	prize_pool	admin_id	status_id	problem_statement	judge_id
H-001	Query Quest	Web development	5/28/2026	6/1/2026	6/3/2026	3000	U-001	1	Perform deep exploratory data analysis tc	U-026
H-002	Malware Mania	Web development	2/22/2026	2/26/2026	2/28/2026	4000	U-003	2	Create a tool for real-time threat detectio	U-026
H-003	Ansible Ace	DevOps	6/5/2026	6/9/2026	6/11/2026	5000	U-021	3	Automate CI/CD pipelines and infrastructu	U-036
H-004	Angular Arena	Web development	5/25/2026	6/1/2026	6/3/2026	4000	U-022	4	Develop a high-performance, responsive v	U-026
H-005	DeFi Derby	Blockchain	5/20/2026	5/24/2026	5/26/2026	5000	U-024	5	Build a decentralized application (dApp) fc	U-038
H-006	ChatBot Championshi	AI & ML	3/1/2026	3/5/2026	3/7/2026	3000	U-025	6	Develop a predictive model or generative	U-030
H-007	Neural Network Ninja	AI & ML	4/10/2026	4/14/2026	4/16/2026	5000	U-002	7	Develop a predictive model or generative	U-029
H-008	NFT Knights	Blockchain	4/27/2026	5/1/2026	5/3/2026	4000	U-004	8	Build a decentralized application (dApp) fc	U-039
H-009	Jenkins Jive	DevOps	4/27/2026	5/1/2026	5/3/2026	4500	U-007	9	Automate CI/CD pipelines and infrastructu	U-040
H-010	Chain Reaction	Blockchain	4/5/2026	4/9/2026	4/11/2026	6000	U-009	10	Build a decentralized application (dApp) fc	U-034
H-011	React Race	Web Development	4/27/2026	5/1/2026	5/3/2026	3000	U-010	11	Develop a high-performance, responsive v	U-041
H-012	Predictive Pros	AI & ML	4/27/2026	5/1/2026	5/3/2026	5500	U-013	12	Develop a predictive model or generative	U-042
H-013	Full Stack Fever	Web Development	4/27/2026	5/1/2026	5/3/2026	5000	U-014	13	Develop a high-performance, responsive v	U-027
H-014	Cloud Castles	Cloud Computing	4/2/2026	4/6/2026	4/8/2026	5000	U-015	14	Design a scalable, serverless architecture t	U-028
H-015	Pipeline Pro	DevOps	4/27/2026	5/1/2026	5/3/2026	6500	U-016	15	Automate CI/CD pipelines and infrastructu	U-036
H-016	Pandas Play	Data Science	4/20/2026	4/24/2026	4/26/2026	4000	U-018	16	Perform deep exploratory data analysis tc	U-032

3:

```sql
ALTER TABLE team_members
ADD COLUMN status VARCHAR(20) DEFAULT 'Pending';
```

team_id	member_id	status
T-001	U-068	Accepted
T-001	U-071	Accepted
T-001	U-077	Accepted
T-002	U-069	Accepted
T-002	U-072	Accepted
T-002	U-078	Accepted
T-003	U-070	Accepted
T-003	U-079	Accepted
T-003	U-081	Accepted
T-004	U-070	Accepted
T-004	U-073	Accepted
T-004	U-076	Accepted
T-005	U-074	Accepted
T-005	U-082	Accepted
T-005	U-083	Accepted
T-006	U-075	Accepted
T-006	U-081	Accepted
T-006	U-084	Accepted
T-007	U-079	Accepted
T-007	U-080	Accepted
T-007	U-085	Accepted
T-008	U-082	Accepted
T-008	U-085	Accepted
T-008	U-086	Accepted
T-009	U-068	Accepted
T-009	U-069	Accepted
T-010	U-069	Accepted
T-010	U-070	Accepted
T-010	U-071	Accepted
T-011	U-070	Accepted
T-011	U-072	Accepted
T-011	U-073	Accepted
T-012	U-071	Accepted
T-012	U-074	Accepted

4.

ALTER TABLE team ADD COLUMN admin_approval VARCHAR(20) DEFAULT 'Pending';

SET SQL_SAFE_UPDATES = 0;

-- Team 1:

```
UPDATE team

SET

    admin_approval = 'Accepted'

WHERE team_id = 'T-001';
```

-- Team 2:

```
UPDATE team
```

```
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-002';
```

-- Team 3:

```
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-003';
```

-- Team 4:

```
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-004';
```

-- Team 5:

```
UPDATE team
SET
    admin_approval = 'Accepted'
WHERE team_id = 'T-005';
```

-- Team 6:

```
UPDATE team

SET

    admin_approval = 'Accepted'

WHERE team_id = 'T-006';
```

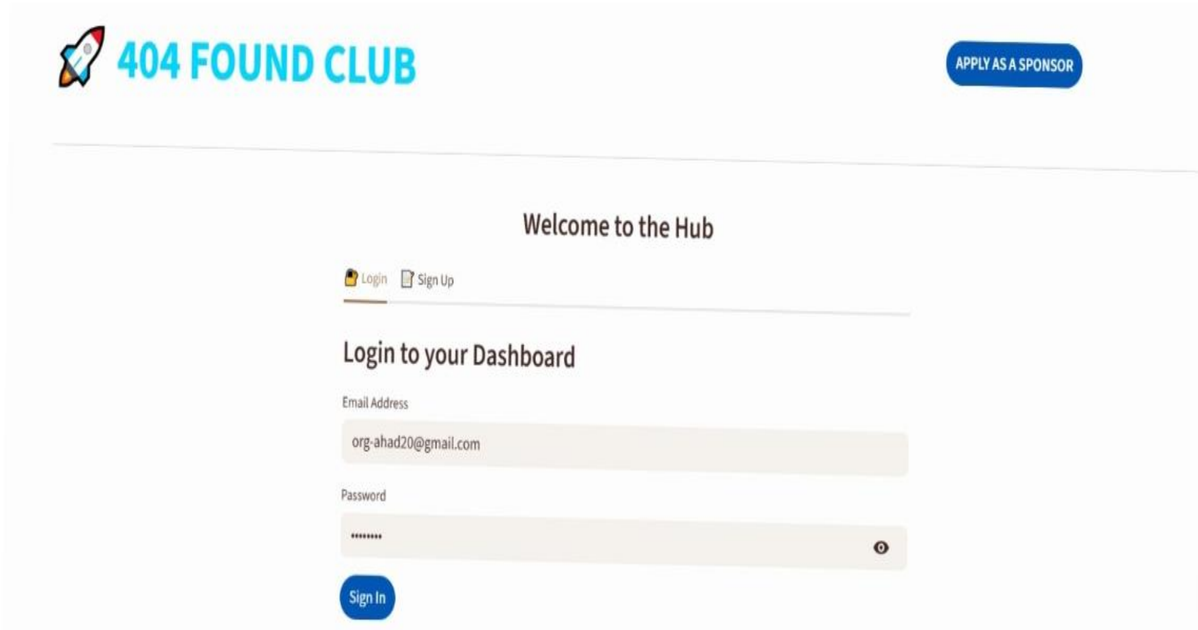
-- Team 7:

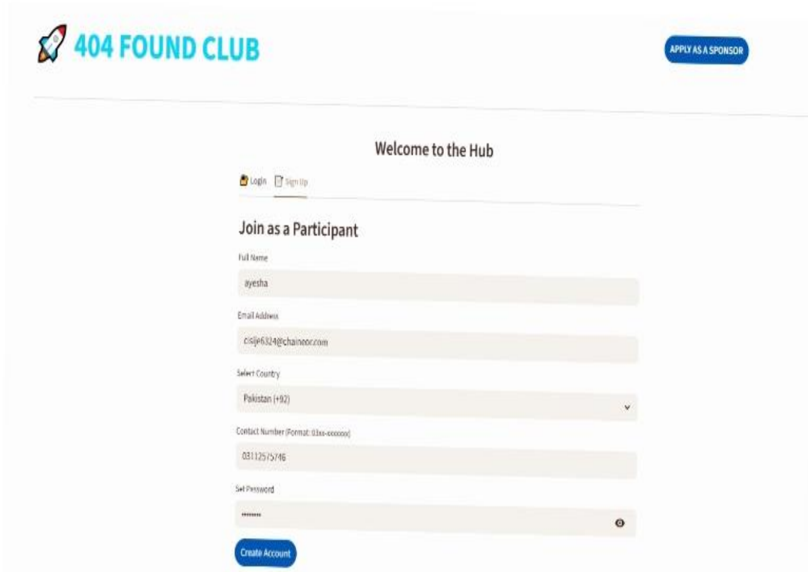
```
UPDATE team

SET

    admin_approval = 'Accepted'

WHERE team_id = 'T-007';
```

team_id	team_name	lead_id	hackathon_id	registration_status	admin_approval
T-001	DataWizards	U-068	H-016	1	Accepted
T-002	SecureSquad	U-069	H-018	1	Accepted
T-003	ByteBusters	U-070	H-006	1	Accepted
T-004	ChainGang	U-073	H-010	1	Accepted
T-005	AppArtisans	U-074	H-017	1	Accepted
T-006	WebWarriors	U-075	H-003	1	Accepted
T-007	CloudKings	U-079	H-019	1	Accepted
T-008	meshing	U-086	H-023	1	Accepted
T-009	NeuralKnights	U-068	H-016	1	Accepted
T-010	VectorVoyagers	U-069	H-016	1	Accepted
T-011	CipherSync	U-070	H-018	1	Accepted
T-012	ZeroTrustZenith	U-071	H-018	1	Accepted
T-013	PromptProphets	U-072	H-006	1	Accepted
T-014	InsightInquisitor	U-073	H-006	1	Accepted
T-015	LogicLoom	U-074	H-006	1	Accepted
T-016	HashHustlers	U-075	H-010	1	Accepted
T-017	BlockBastions	U-076	H-010	1	Accepted
T-018	BinaryBridges	U-077	H-010	1	Accepted
T-019	SwiftSynthetics	U-078	H-017	1	Accepted
T-020	ServerlessSamur	U-080	H-019	1	Accepted
T-021	DataDrifters	U-082	H-008	1	Accepted

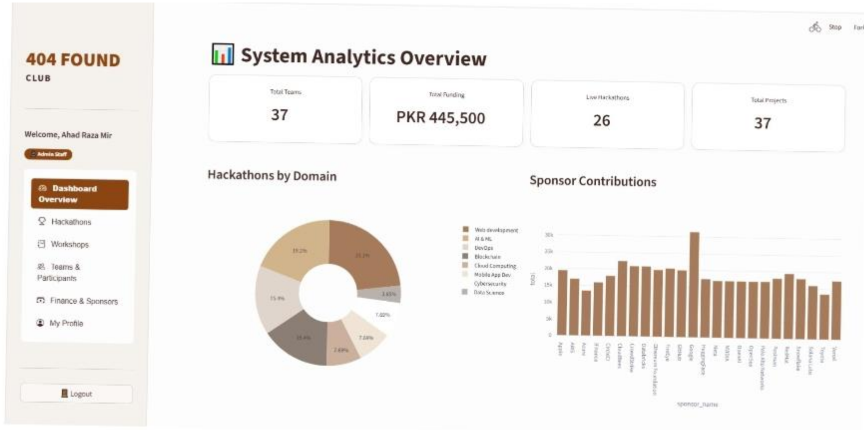
6. User Interface UI:

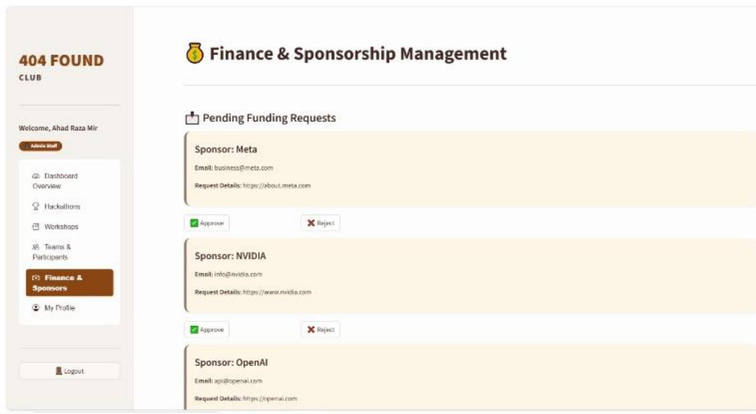
LOGIN/SIGNUP PAGE:



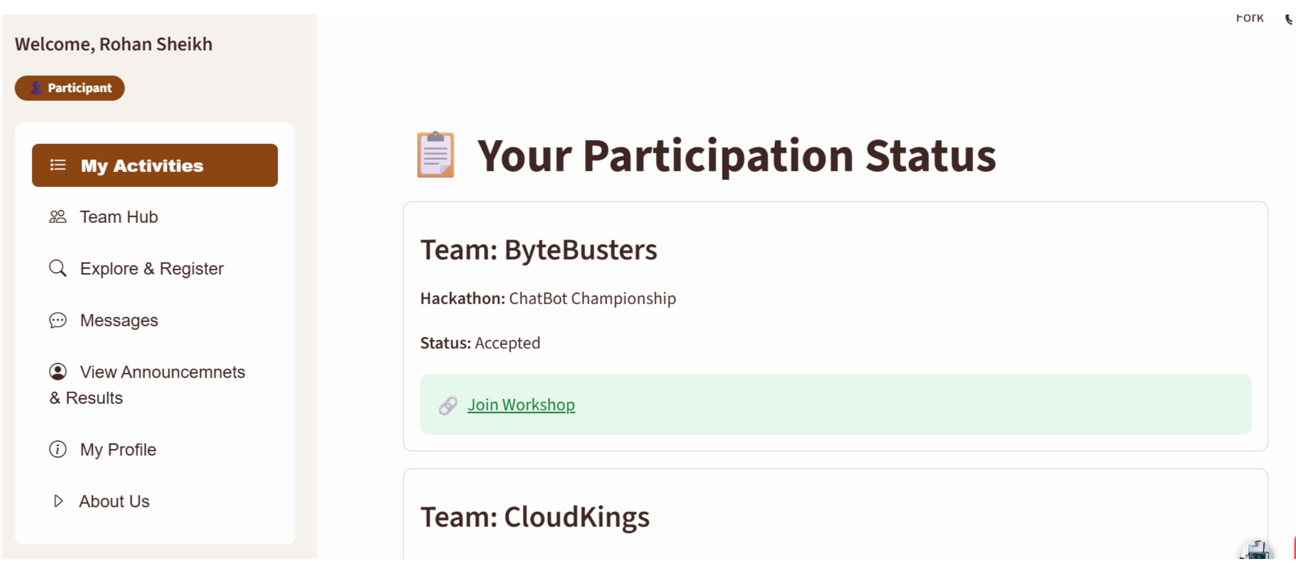


ADMIN DASHBOARD:





PARTICIPANT DASHBOARD:



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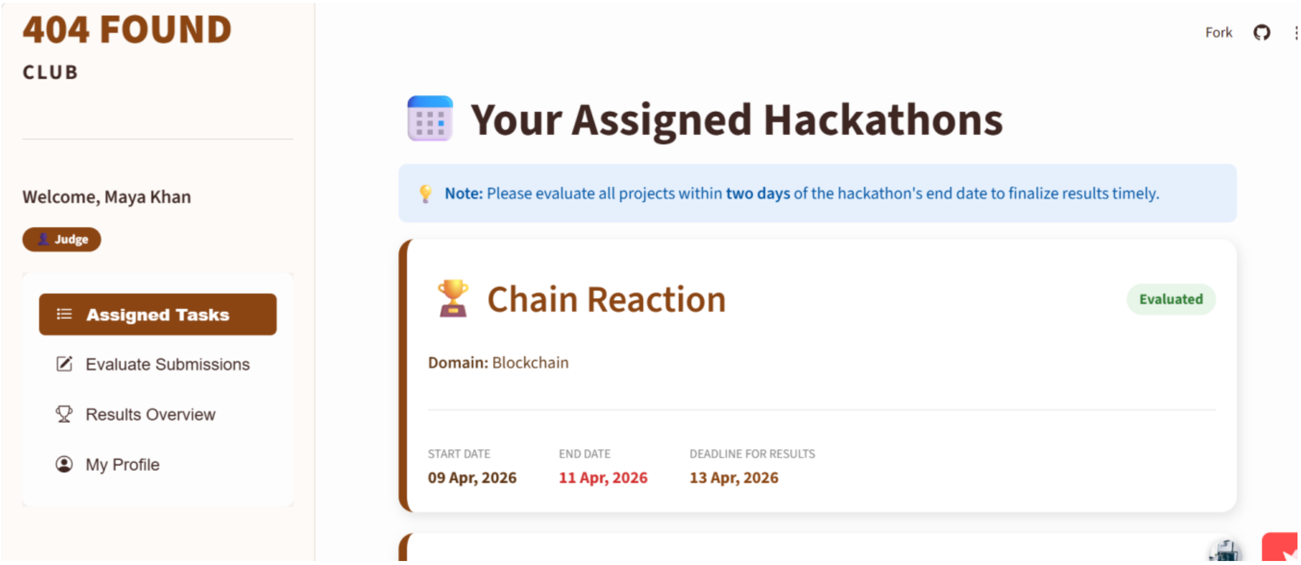
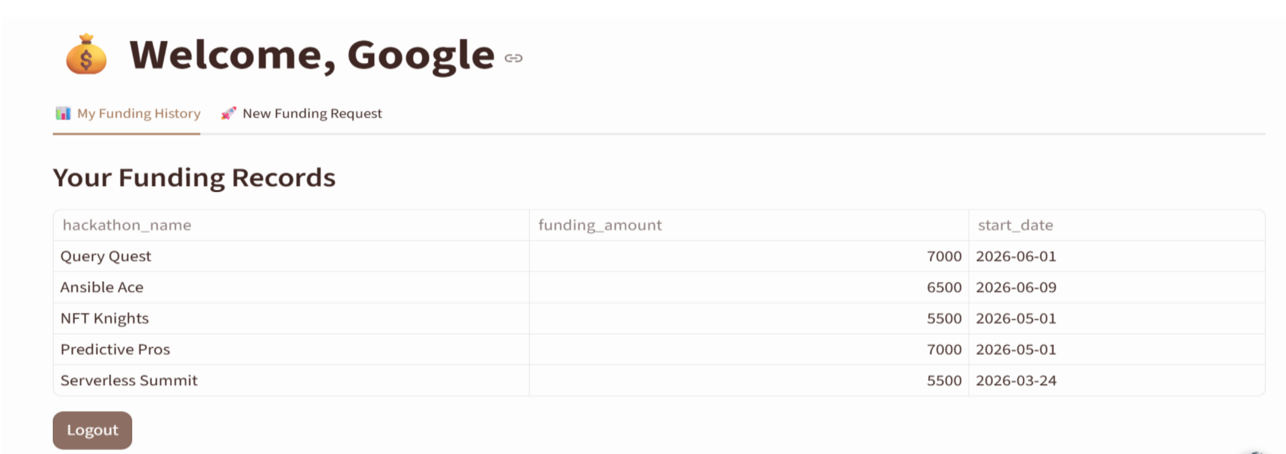


MENTOR DASHBOARD:

SPONSER DASHBOARD:



6. Conclusion:

The 404 Found Club Hackathon Management System transitions a fragmented workflow into a structured, scalable SQLite platform. By progressing from requirements analysis to BCNF normalization, the system eliminates redundancies and anomalies across seven core modules. Using junction tables for many-to-many relationships and enforcing strict referential integrity, the design ensures data consistency for four distinct user roles, resulting in a maintainable and efficient management tool.

TEAM PARTICIPATION

The following roles and responsibilities were assigned among team members for the successful completion of the project:

UI Designing: Izma Warsia, Safa Khurram

Normalization and Report Writing: Safa Khurram, Saniya Siddiqui, Ayesha

Physical Model Development: Izma Warsia

Research Work: Safa Khurram, Ayesha

Data Fetching and Handling: Izma Warsia, Saniya Siddiqui

Entity Relationship Diagram (ERD / EERD) : Ayesha, Safa Khurram, Saniya Siddiqui,